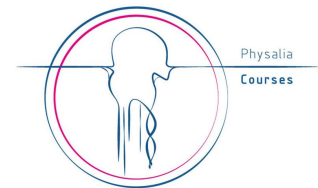


Course: Fundamentals of Biostatistics with R

Eliana Ibrahimi
Associate Professor in Biostatistics
University of Tirana, Albania



Welcome

Get to know each other

Your name? Your position?

What field or background do you come from?

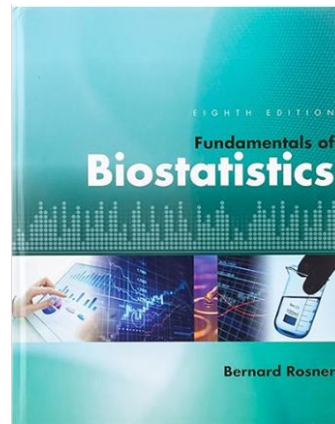
Your experience with data analysis & R.

About this course

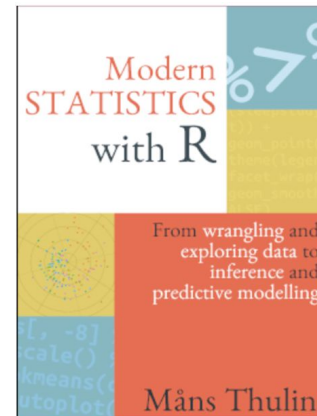
- This is an introductory-level biostatistics course and is designed for researchers and students interested in medicine or other health-related areas.
- It requires no previous background in statistics, and its mathematical level assumes only a knowledge of algebra.

Resources/Learning materials

- BiostatCourse lectures
- R Notebooks (rmd and html format)
- Fundamentals of Biostatistics, 8th Edition
- Modern Statistics with R
- Online tutorials on software applications
- Claude AI to support coding.



[Fundamentals of Biostatistics 8th Edition](#)



<https://www.modernstatisticswithr.com/>

Activities

- Lectures (Course handouts)
- Guided hands-on sessions (R Notebooks)
- Supervised Open lab activities (Small practical challenges)

Note: Materials available in Drive, small updates can be done during the course.

Course outline

Day 1

Intro to biostat

Intro to biostatistics and
medical data

Intro to R and data
exploration/management

Day 2

Probability, Sampling, CIs

Probability concepts and
distributions

Sample size and power
calculations, concepts of
hypothesis testing

Day 3

Univariate statistical analysis and linear models

Parametric and
non-parametric tests

Linear models

Course outline

Day 4

Categorical data analysis

Chi-square and Fisher exact tests

Generalized linear models (logistic and Poisson regression)

Day 5

Survival analysis

Intro to survival analysis

Kaplan-Meier Estimator,
Cox Proportional hazard model

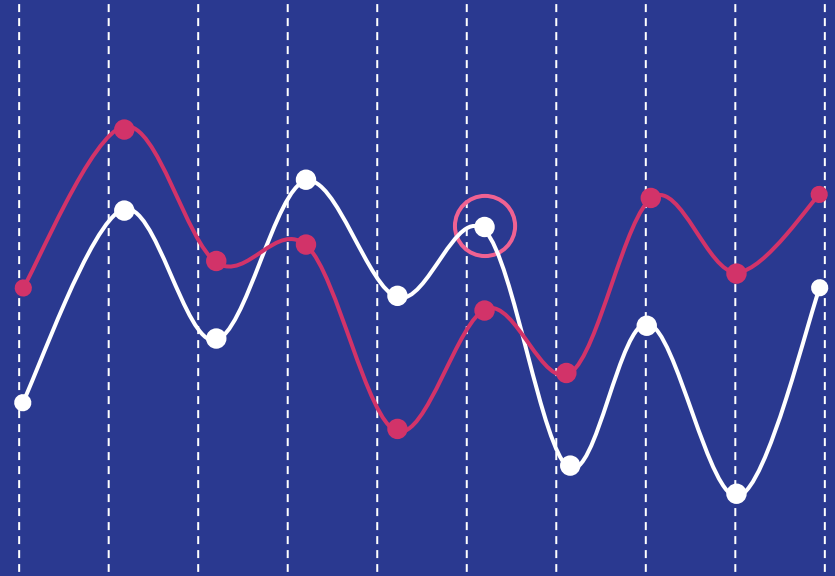
Documentation & reproducibility

Post course

Hands-on Project

Analyze a dataset, prepare and submit a report to get feedback from the instructor.

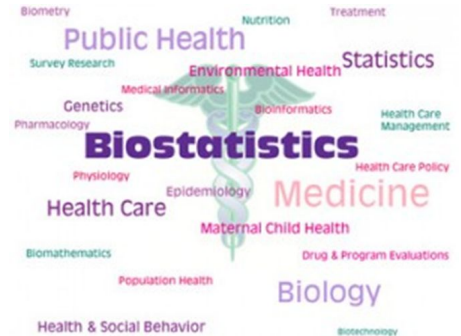
Lecture 1: Intro to Biostatistics



What is Biostatistics?

Biostatistics is a branch of applied statistics that applies statistical methods to collect, analyze, and interpret data related to biology, public health, and medicine.

--- Much more statistics than medicine/biology, however biostatisticians must learn the biology also.



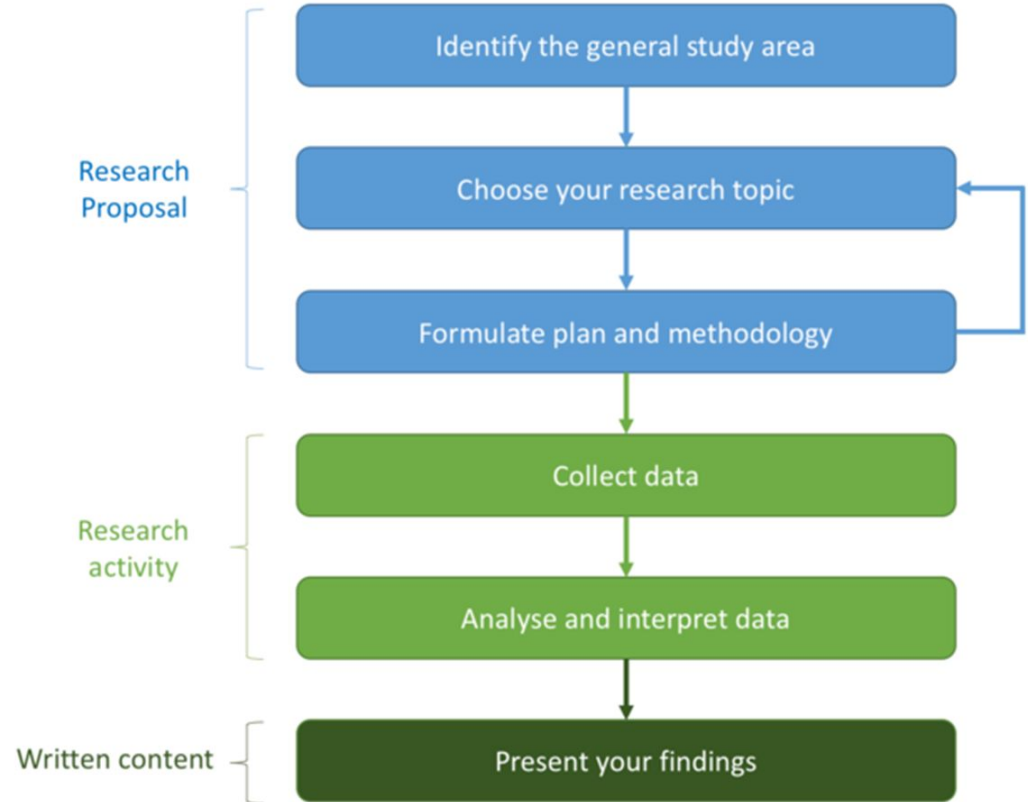
Why do I need to learn Biostatistics?

- To effectively conduct research
- To be able to read scientific papers
- To further develop critical and analytical thinking



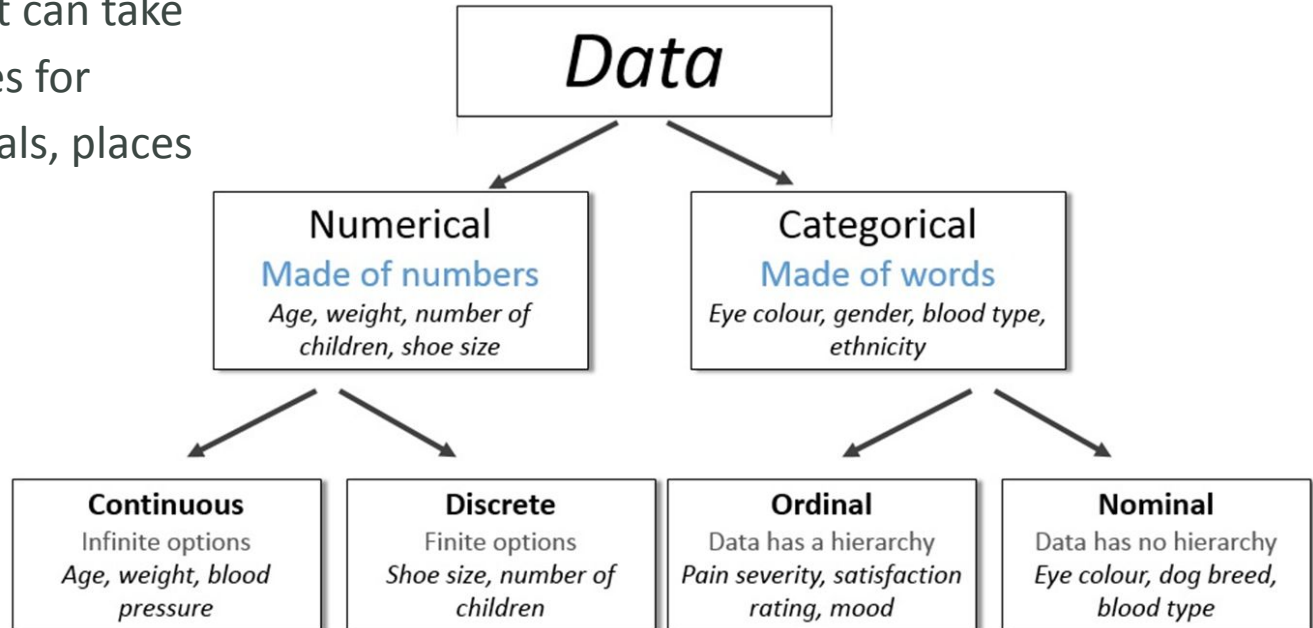
Role of Biostatistics in Research

The biostatistician should be present in each step.



Biostatistics deals with data (variables)

- A random variable is a characteristic that can take on different values for different individuals, places or things.

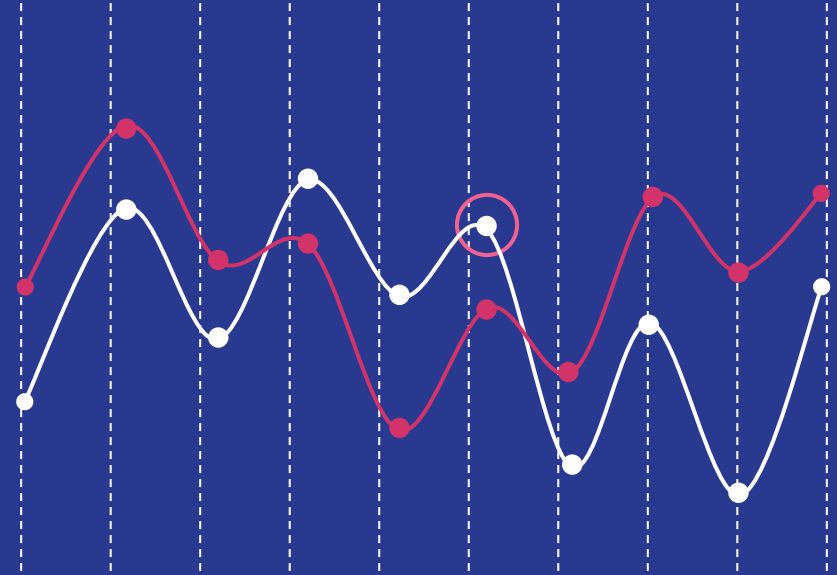


Class activity

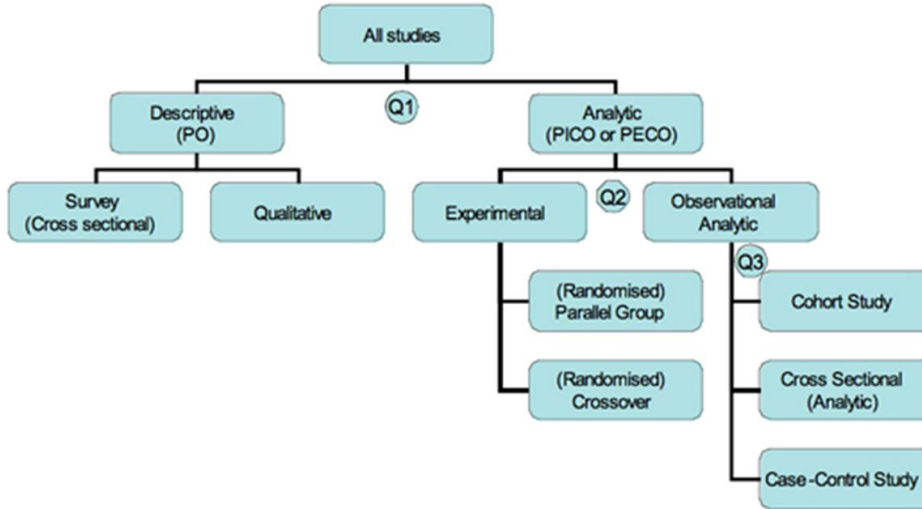
Identify the type of variable

- Heart rate (beats per minute) _____
- Disease stage (stage I, II, III, IV) _____
- Body temperature _____
- Number of asthma attacks _____
- Smoking status _____
- Cholesterol level _____

Lecture 2: Data collection - Study Design



Data collection/Study design



PO=Population + Outcome

PICO=Population + Intervention + Comparator + Outcome

PECO=Population + Exposure + Comparator + Outcome

Q1. What was the aim of the study?

Q2. If analytic, was the intervention randomly allocated?

Q3. When were the outcomes determined?



Reading

Data collection/Study design

Experimental

Researcher assigns exposure or treatment.
Can establish causation.

Random assignment to treatment or control; outcome measured after.

- Drug A vs. placebo on blood pressure in hypertensive patients.

Observational

Researcher observes without intervening. Can establish association.

Follow exposed and unexposed groups forward in time and compare.

- Smokers vs. non-smokers followed 10 years for lung cancer incidence.

Class activity

A researcher wants to know whether daily coffee consumption increases the risk of atrial fibrillation. She has two options:

- **Option A:** Recruit 500 volunteers, randomly assign half to drink 3 cups of coffee per day for 6 months, and monitor for arrhythmia events.
- **Option B:** Survey 2,000 adults about their coffee habits and medical history, then compare arrhythmia rates between heavy and light drinkers.

What study design does each option represent?

Bias and confounding

A study can be large, well-executed, and still produce the wrong answer. Bias and confounding are systematic errors that don't cancel out with larger samples.

Selection bias

Who ends up in the study is not representative of the target population.

Information bias

Exposure or outcome is measured incorrectly or differently across groups.

Confounding

A third variable is associated with both exposure and outcome, distorting the apparent effect.

In depth examples

Selection: Loss to follow-up

Participants who drop out of a cohort differ systematically from those who remain, skewing the outcome estimate.

Sickest patients in a drug trial discontinue early due to side effects; remaining completers look artificially healthier.

Information: Recall bias

Cases recall past exposures more thoroughly than controls because they have searched for an explanation for their illness.

Mothers of children with birth defects report more medication use in pregnancy than mothers of healthy children — partly because they have thought harder about it.

Confounding: Classic confounder

A variable associated with both exposure and outcome that is not on the causal pathway — it creates a spurious or masked association.

Coffee drinkers smoke more. Observed coffee–lung cancer association is partly (or wholly) driven by smoking, not coffee.

Software



Currently widely used software are SAS, SPSS, Stata, **R**, Python, MINITAB, and Excel.

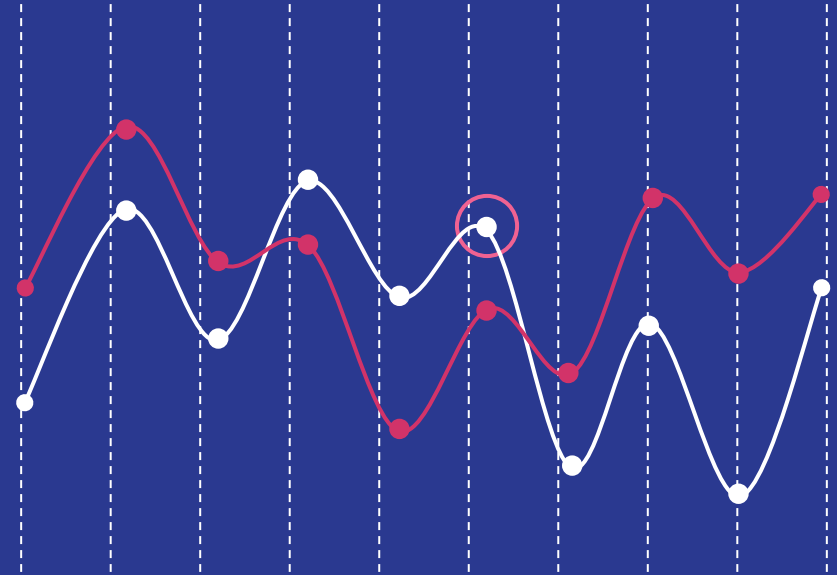
Why R?

- Free and open-source, no licensing costs.
- Rich statistical libraries built for biostatistical analysis and visualization.
- Reproducibility, R scripts document every step of an analysis, making studies easier to replicate and audit.
- Active community — a large network of biostatisticians and epidemiologists continuously contribute packages, tutorials, and peer support.

Hands-on activity

R Notebook: Hands-on1_Intro to R

Lecture 3: Data Exploration



Data analysis

Data analysis is a process of cleaning, transforming, exploring, and modeling data to discover useful information and supporting decision-making.



Data preparation

Data preparation includes editing, coding, data entry and is the process that ensures data accuracy and their transformation from raw to organized forms that are appropriate for analysis.

Data Exploration: Categorical Variables

Frequency tables

Count (How frequent values occur)

Relative (The % of observations with a specific value)

Graphs (simple and clustered)

Bar

Pie

Area

Data Exploration: Numerical Variables

Descriptive statistics

Mean, Mode, Median

Variance, Standard deviation, SE

Range, Percentiles

Graphs (simple and clustered)

Histogram

Boxplot

Error bars

Scatterplot

...



Descriptive Statistics

Measures of central tendency

1. Mean – the arithmetic average of all values in a dataset
2. Median – the middle value when data is arranged in order
3. Mode – the most frequently occurring value in a dataset

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

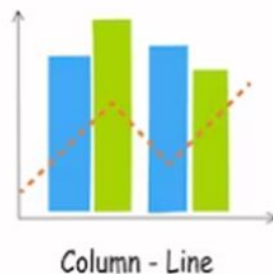
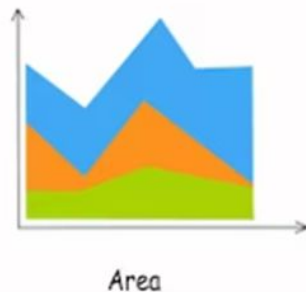
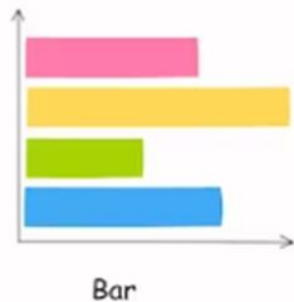
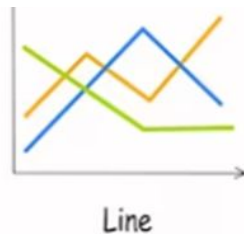
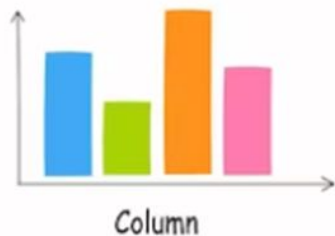
Measures of spread

4. Range – the difference between the maximum and minimum values
5. Variance – the average of squared deviations from the mean, how spread are the data
6. Standard deviation – the square root of variance, same units as the data
7. Interquartile range (IQR) – the difference between the 75th and 25th percentiles, capturing the middle 50% of data

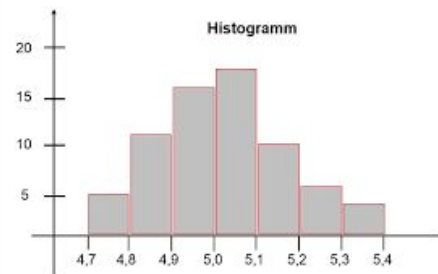
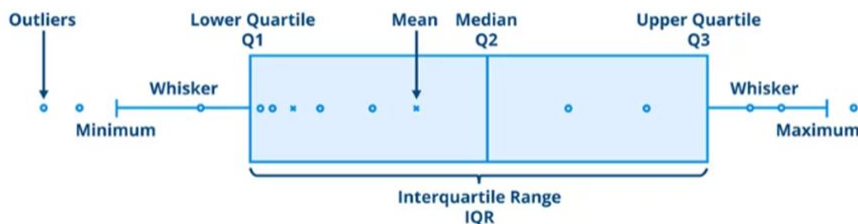
$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}$$

$$s = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}}$$

Graphs



Box plot

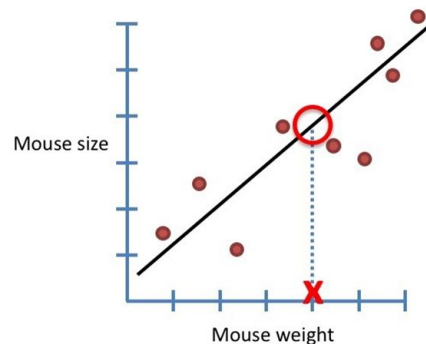


Statistical modeling

A statistical model is usually specified as a mathematical relationship between variables and represents, often in considerably idealized form, the data-generating process.

Includes

T-tests, ANOVA, Linear Regression, Generalized linear models, and many more...



Draw conclusions

While interpreting the results ask yourself:

- Did the analysis answer my research question?
- Was there any limitation in my analysis which would affect my conclusions?
- Was the analysis sufficient enough to help decision making?

Communicate the findings

- Documentation of statistics
- Making presentations
- Writing reports, or blogs
- Writing manuscripts

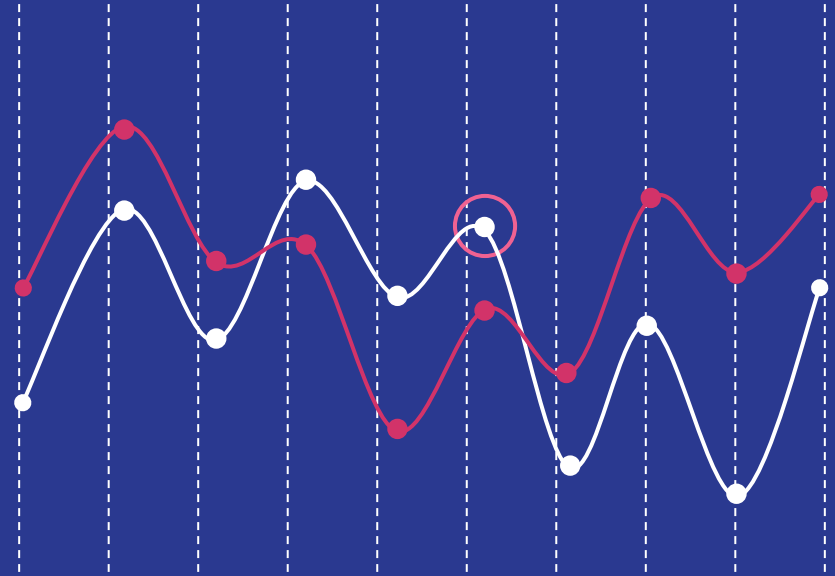
Extra skills needed at this stage

- Writing
- Presenting
- Communication

Hands-on Activity

R Notebook: Hands-on2_Data Exploration in R

Lecture 4: Probability and Distributions



Learning objectives

- Apply core probability rules to interpret clinical data
- Explain conditional probability, sensitivity, specificity, and predictive value
- Identify the right distribution for a given type of outcome
- Interpret Bayes' theorem intuitively using a diagnostic scenario

A clinical puzzle

- A rare disease affects 1% of the population. A diagnostic test has a **95% sensitivity** (true positive rate) and a **95% specificity** (true negative rate). A patient tests positive.

What is the probability they actually have the disease?

Definition of probability

- **Frequentist**

The probability of an event is the long-run relative frequency of the set of outcomes over an indefinitely large (or infinite) number of trials.

- **Bayesian**

Probability = degree of belief, updated with evidence.

Useful when we can't repeat an experiment (e.g., disease in a patient).



Key rules of probability

Addition Rule

$P(A \text{ or } B) = P(A) + P(B)$ (for mutually exclusive events)

$P(\text{blood type A or B}) = P(A) + P(B)$

Multiplication Rule

$P(A \text{ and } B) = P(A) \times P(B)$ (for independent events)

$P(\text{two positive tests}) = 0.05 \times 0.05 = 0.0025$

Complement Rule

$P(A) = 1 - P(\text{not } A)$ (always true)

$P(\text{disease}) = 1 - P(\text{no disease})$

Conditional probability

$P(A \mid B)$ = Probability of A, given that B has occurred

Sensitivity

$P(\text{positive test} \mid \text{disease})$

How well the test detects true cases?

Specificity

$P(\text{negative test} \mid \text{no disease})$

How well does the test rule out disease?

PPV

$P(\text{disease} \mid \text{positive test})$

If positive, what's the chance of disease?

NPV

$P(\text{no disease} \mid \text{negative test})$

If negative, what's the chance of no disease?

Bayes Theorem

How to update the probability of disease given new evidence (a test result)?

$$P(\text{Disease} \mid \text{Positive test}) = \frac{P(\text{Positive} \mid \text{Disease}) \times P(\text{Disease})}{P(\text{Positive})}$$

$P(\text{Positive})$ is rarely known directly, so it is expanded as:

$$P(\text{Positive}) = P(\text{Positive} \mid \text{Disease}) \times P(\text{Disease}) + P(\text{Positive} \mid \text{No Disease}) \times P(\text{No Disease})$$

where $P(\text{Positive} \mid \text{No Disease}) = 1 - \text{Specificity}$ (the false positive rate).

Bayes Theorem

In clinical terms:

$$\text{PPV} = \frac{\text{Sensitivity} \times \text{Prevalence}}{\text{Sensitivity} \times \text{Prevalence} + (1 - \text{Specificity}) \times (1 - \text{Prevalence})}$$

Solve the clinical puzzle

- A rare disease affects 1% of the population. A diagnostic test has a **95% sensitivity** (true positive rate) and a **95% specificity** (true negative rate). A patient tests positive.

What is the probability they actually have the disease?

A) 95%, B) 75%, C) 50%, or D) Less than 20%.

Binomial distribution

- Binary outcome (success/failure, yes/no)
- Fixed number of trials (n)
- Each trial has the same probability (p)
- Trials are independent

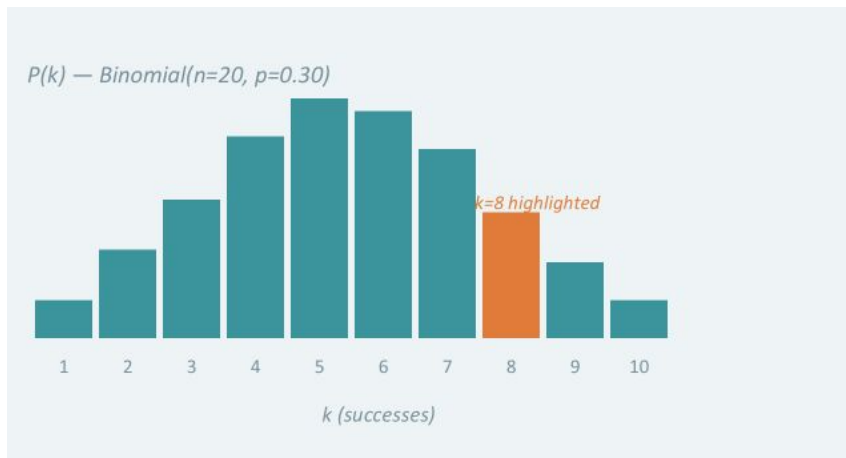
$$Pr(X = k) = \binom{n}{k} p^k q^{n-k}, \quad k = 0, 1, \dots, n$$

Example: In a trial of 20 patients, if response rate is 30%, what is the probability that exactly 8 respond?

n = number of trials (20 patients)

p = probability of success per trial (0.30)

Mean = np Variance = $np(1-p)$



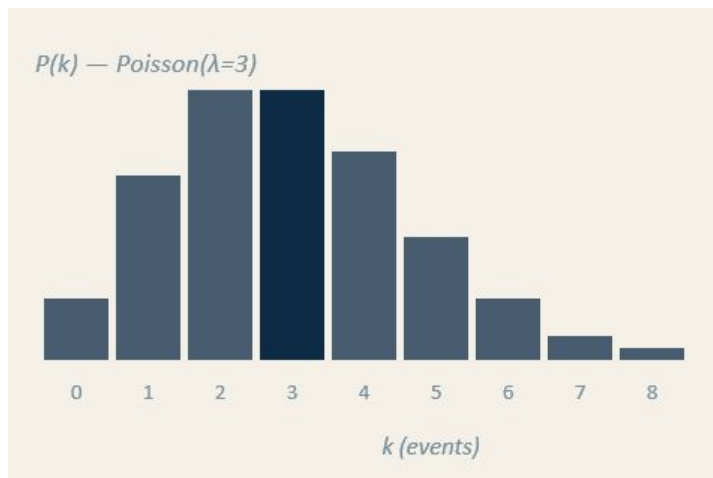
Poisson distribution

- Counting rare events, over a fixed time or space
- Events occur independently, rate is constant

Example: Hospital-acquired infections per month
Adverse drug reactions per 1,000 doses
Emergency admissions per day

$$Pr(X = k) = e^{-\mu} \mu^k / k!, \quad k = 0, 1, 2, \dots$$

where $\mu = \lambda t$ and e is approximately 2.71828.

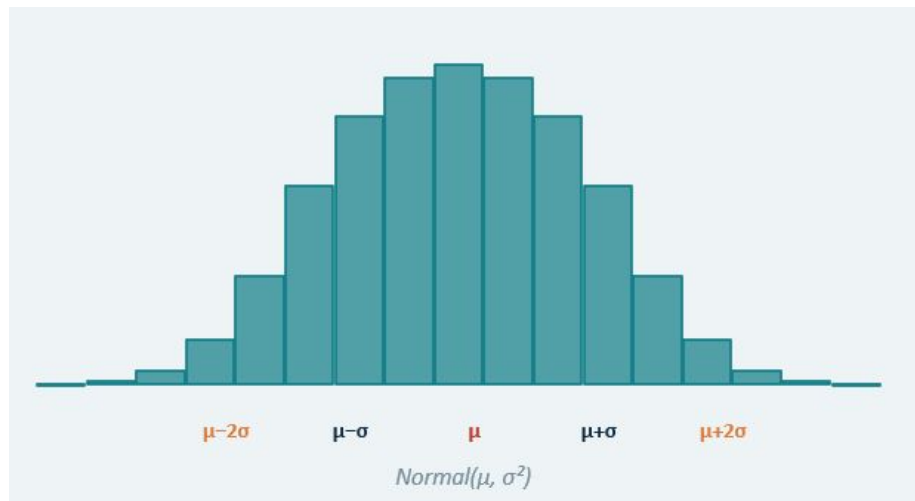


Normal distribution

- **Symmetric bell shape;**
defined by μ (mean) and σ (SD)

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{1}{2\sigma^2}(x-\mu)^2\right], \quad -\infty < x < \infty$$

for some parameters μ , σ , where $\sigma > 0$.



Log-Normal distribution

- **Data is skewed right (positive)**

- **Values must be > 0**

The log of the variable is normally distributed

- **Clinical examples**

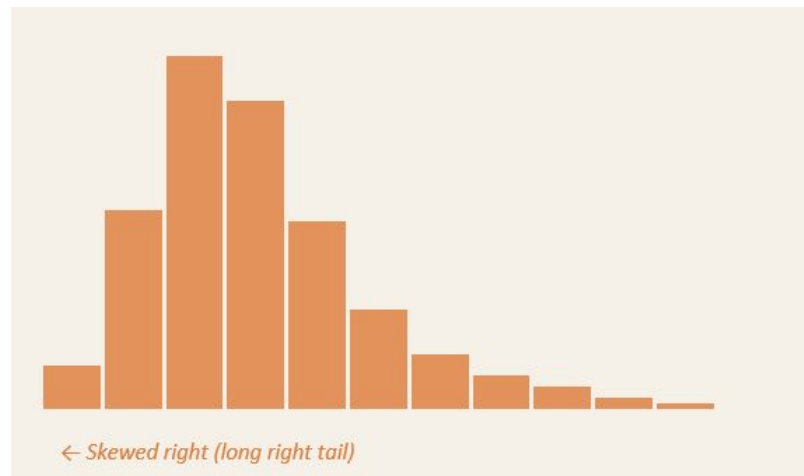
Viral loads, enzyme levels,
survival times, drug concentrations

- **Practical implication**

Report geometric mean

Use log scale for plots

Log-transform before analysis



Take $\log(X)$ → approximately Normal

Choosing the wrong distribution leads to wrong inference

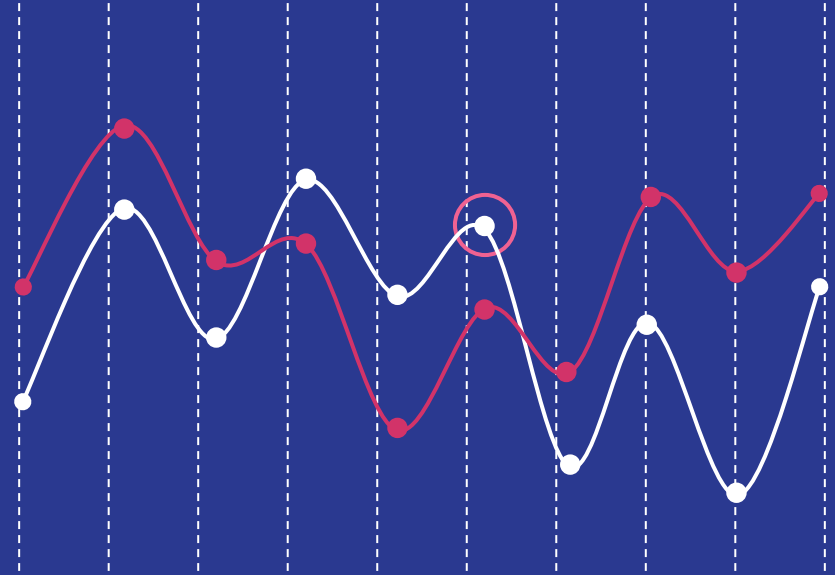
Outcome Type	Clinical Example	Distribution
Binary (yes/no)	<i>Responded to treatment?</i>	Binomial
Count of rare events	<i>Infections per month</i>	Poisson
Continuous, symmetric	<i>Blood pressure, height</i>	Normal
Continuous, right-skewed, >0	<i>Viral load, enzyme level</i>	Log-Normal
Continuous, small sample	<i>Mean from $n < 30$</i>	t-distribution

Hands-on Activity

R Notebook: Hands-on 3_Probability distributions
and Bayes theorem in R

Lecture 5:

Sampling basics & Confidence Intervals



Sampling: Learning objectives

- Define population, sample, parameter, statistic, sampling frame
- Identify and explain the four probability sampling methods
- Compare probability vs. non-probability sampling on bias and generalizability
- Recognize selection bias in a scenario
- Select an appropriate sampling strategy and justify it

Why we sample?

"How did researchers know that smoking causes lung cancer before studying the entire population?"

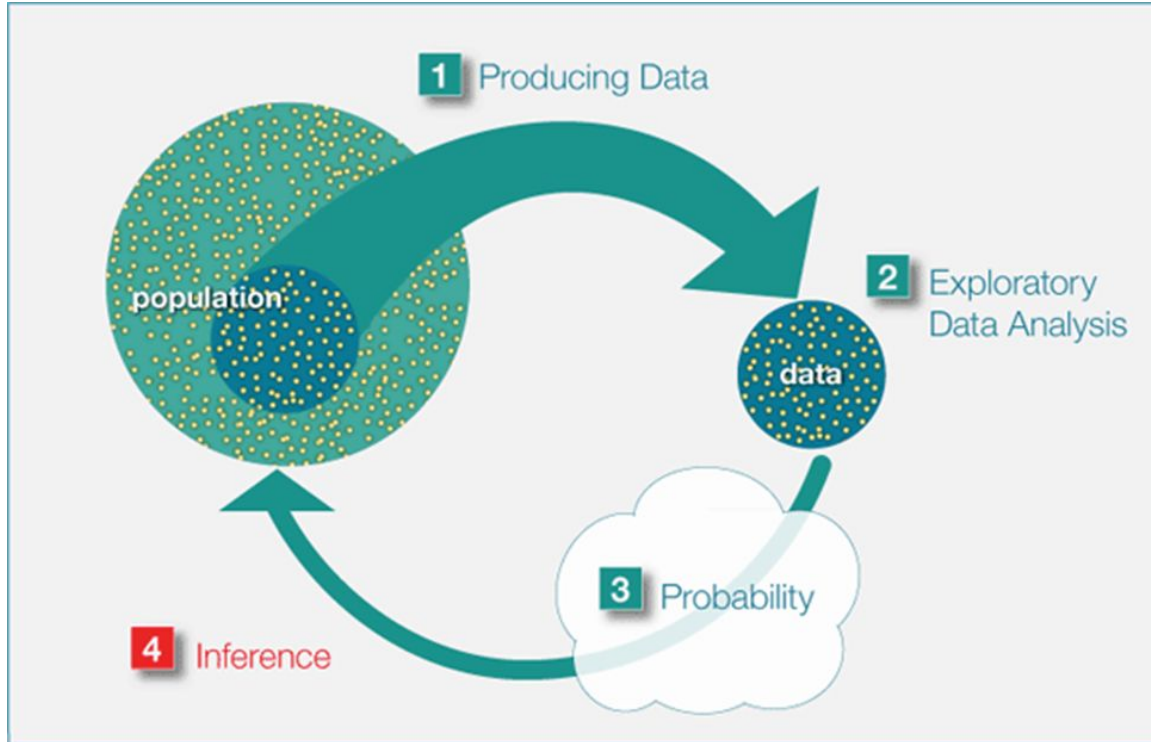
We want to know something about everyone, but we can only study a few...

Why we sample?

- Populations are too large, dispersed, or expensive to enumerate
- Some measurements are destructive or invasive
- A well-drawn sample can be more accurate than a rushed census

Goal: make valid inferences about a population from a subset

Aim of Statistical Inference



- Draw conclusions about a population based on the data obtained from a sample chosen from it.

Key definitions

- **Target population** — everyone we want to draw conclusions about
- **Study population** — the accessible subset we can actually reach
- **Sample** — the units we actually measure
- **Sampling frame** — the list or mechanism used to select the sample

Parameters vs Statistics

	Population	Sample
Mean	μ (parameter)	$\bar{x}$ (statistic)
Proportion	p (parameter)	$\hat{p}$ (statistic)

We observe statistics. We use them to *estimate* parameters.

Hypertension Prevalence

What proportion of adults in a city have hypertension?

- Target population: all adults in the city (~500,000)
- Study population: adults registered at primary care clinics
- Sampling frame: clinic patient lists
- Sample: 800 randomly selected patients

What are the risks of using clinic lists as the frame?

Probabilistic sampling methods

Simple Random Sampling

Every unit has an equal probability of selection.

- Use a random number table or software
- Unbiased and easy to analyze statistically
- Requires a complete sampling frame
- Can be costly if population is geographically dispersed

Best for: small, well-defined, accessible populations

Systematic Sampling

Select every k -th unit from an ordered list. k = population size / desired sample size.

- Fast and easy to implement in the field
- Watch out for periodicity — if the list has a hidden pattern at interval k , bias is introduced

Example: every 10th patient admitted to a ward

Stratified Sampling

Divide the population into strata, then sample within each stratum.

- Strata: age group, sex, region, disease severity
- Ensures all subgroups are represented
- Improves precision compared to SRS of the same size

Best for: heterogeneous populations where subgroup estimates matter

Cluster Sampling

Randomly select groups (clusters), then measure all or some units within each cluster.

- Clusters: villages, schools, clinics, households
- Reduces travel and logistical costs dramatically
- Less statistically efficient than SRS — requires design effect adjustment in analysis

Best for: large field surveys in resource-limited settings

Comparing methods

Method	Requires full frame?	Cost	Precision
Simple random	Yes	High	High
Systematic	Yes (ordered)	Medium	High
Stratified	Yes (by strata)	Medium	Higher
Cluster	No	Low	Lower

Non-Probability Sampling

Selection is not based on known, equal probabilities.

- **Convenience** — whoever is easiest to reach
- **Purposive** — selected by researcher judgment
- **Snowball** — participants recruit further participants

Common in qualitative research, pilot studies, and exploratory work. *Cannot calculate standard errors or make formal inferences to a population.*

Class activity

- You want to study the average BMI of patients in a city. You decide to recruit the first 50 patients who walk into a specific cardiology clinic on a Monday morning.

What type of sampling we apply here? "Is this a representative sample of the city's population? Why or why not?"

Selection Bias

Occurs when the sample differs systematically from the target population.

Classic example: 1936 Literary Digest poll predicted a Landon landslide — sampled from telephone directories and car registrations, missing poorer voters who favored Roosevelt.

Ask of every study: who was left out of the sampling frame?

Representativeness

A sample is representative if it reflects the key characteristics of the target population.

- Achieved through good study design, not just large n
- A large biased sample is worse than a small unbiased one
- Determines external validity — whether findings generalize

Sample Size: The Intuition

Larger samples give more precise estimates.

$$SE = \sigma / \sqrt{n}$$

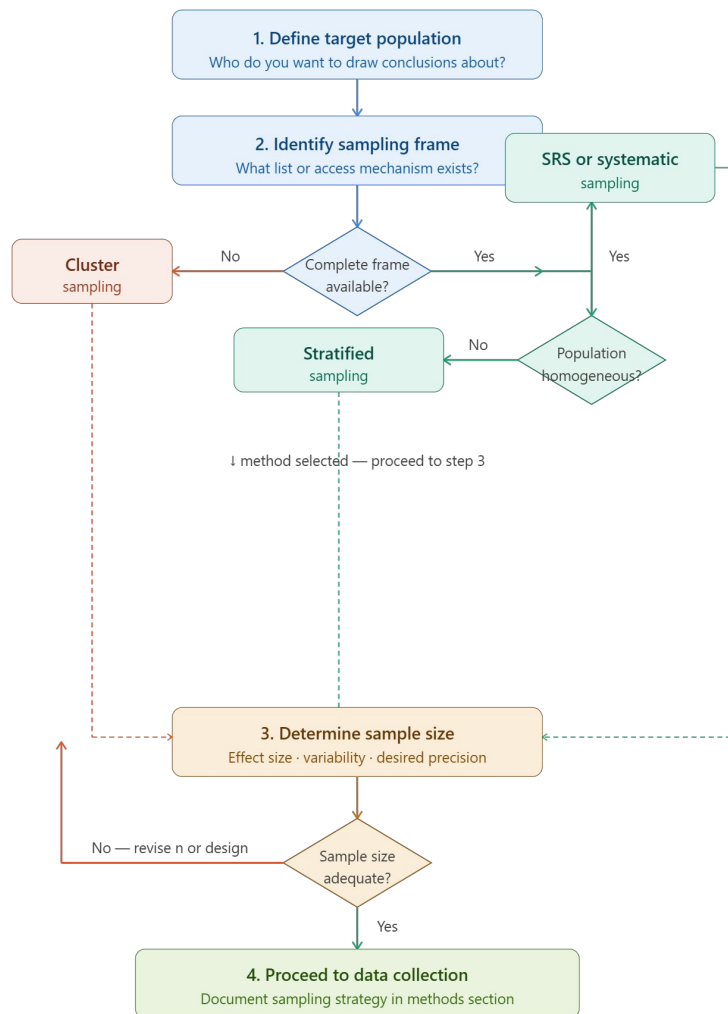
- As n doubles, SE shrinks by a factor of $\sqrt{2}$ (~29%)
- To halve the SE, we need to quadruple n

Three factors that drive required sample size:

1. Expected effect size
2. Variability in the outcome
3. Desired precision (or power)

The Sampling Decision Tree

At every step, ask: who is systematically excluded from my frame or method? That exclusion is where selection bias enters.



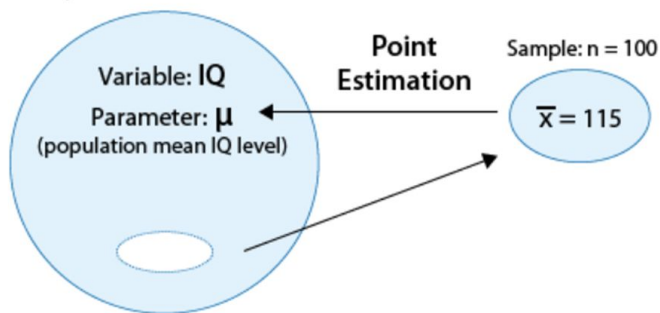
Sampling summary

- Sampling allows valid inference without studying everyone
- Probability methods (SRS, systematic, stratified, cluster) support statistical inference
- Non-probability methods are useful but limit generalizability
- Bias comes from the *design*, not just the size
- Precision improves with n , but design quality matters more

ESTIMATION: Learning objectives

- Distinguish point estimates from interval estimates
- Interpret a 95% confidence interval correctly
- Construct a confidence interval for a mean and a proportion using the standard error and the appropriate z critical value, given sample data
- Explain how sample size, variability, and confidence level each affect the width of a confidence interval

Point estimation



- In **point estimation**, we estimate an unknown parameter using a single number that is calculated from the sample data.
- Point estimates are totally unbiased estimates for the population parameter only if the sample is random and the study design is not flawed.

Interval estimation

- In **interval estimation**, we estimate an unknown parameter using an interval of values that is likely to contain the true value of that parameter.

The diagram illustrates the formula for interval estimation: $\bar{x} \pm T_c \cdot s/\sqrt{n}$. Annotations include:

- Sample Mean and center of interval**: Points to $\bar{x}$.
- Critical T-value (depends on confidence level)**: Points to T_c .
- Standard Error**: Points to $s/\sqrt{n}$.
- Margin of Error**: A bracket under the product $T_c \cdot s/\sqrt{n}$ points to this label.

CIs of the mean



Fundamentals of Biostatistics
Chapter 6

Confidence Interval for the Mean of a Normal Distribution

A $100\% \times (1 - \alpha)$ CI for the mean μ of a normal distribution with unknown variance is given by

$$\left(\bar{x} - t_{n-1, 1-\alpha/2} s / \sqrt{n}, \bar{x} + t_{n-1, 1-\alpha/2} s / \sqrt{n} \right)$$

A shorthand notation for the CI is

$$\bar{x} \pm t_{n-1, 1-\alpha/2} s / \sqrt{n}$$

Confidence Interval for the Mean of a Normal Distribution (Large-Sample Case)

An approximate $100\% \times (1 - \alpha)$ CI for the mean μ of a normal distribution with unknown variance is given by

$$\left(\bar{x} - z_{1-\alpha/2} s / \sqrt{n}, \bar{x} + z_{1-\alpha/2} s / \sqrt{n} \right)$$

Factors affecting the length of CI

Factors Affecting the Length of a CI

The length of a $100\% \times (1 - \alpha)$ CI for μ equals $2t_{n-1, 1-\alpha/2} s / \sqrt{n}$ and is determined by n , s , and α .

- n As the sample size (n) increases, the length of the CI decreases.
- s As the standard deviation (s), which reflects the variability of the distribution of individual observations, increases, the length of the CI increases.
- α As the confidence desired increases (α decreases), the length of the CI increases.

95% cls for proportion

Normal-Theory Method for Obtaining a CI for the Binomial Parameter p (Wald Method)

An approximate $100\% \times (1 - \alpha)$ CI for the binomial parameter p based on the normal approximation to the binomial distribution is given by

$$\hat{p} \pm z_{1-\alpha/2} \sqrt{\hat{p}\hat{q}/n}$$

This method of interval estimation should only be used if $n\hat{p}\hat{q} \geq 5$.

Hypothesis testing

Statistical hypothesis testing is defined as:

Assessing evidence provided by the data against the null hypothesis.

Step 1

- **Formulate the hypotheses:**
- H_0 – null and H_1 – alternative

Step 2

- Collect relevant data and summarize them.

Step 3

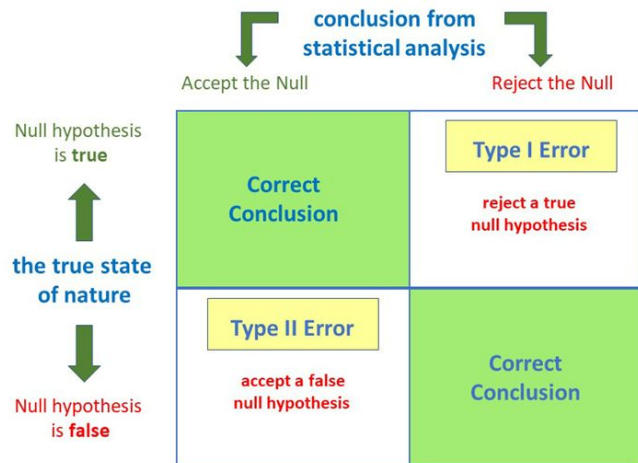
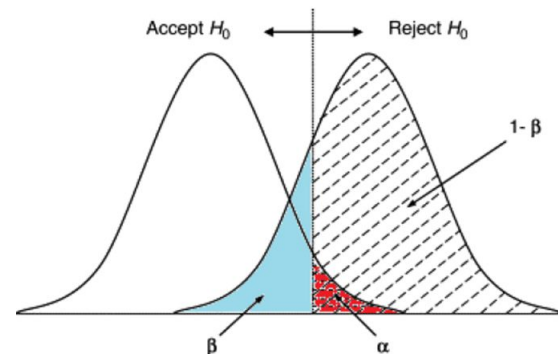
- Test how likely it is to observe data we obtained, if null hypotheses is true. **Compute test statistics**

Step 4

- **Compute p -value** and make our decision.

Type I and II errors

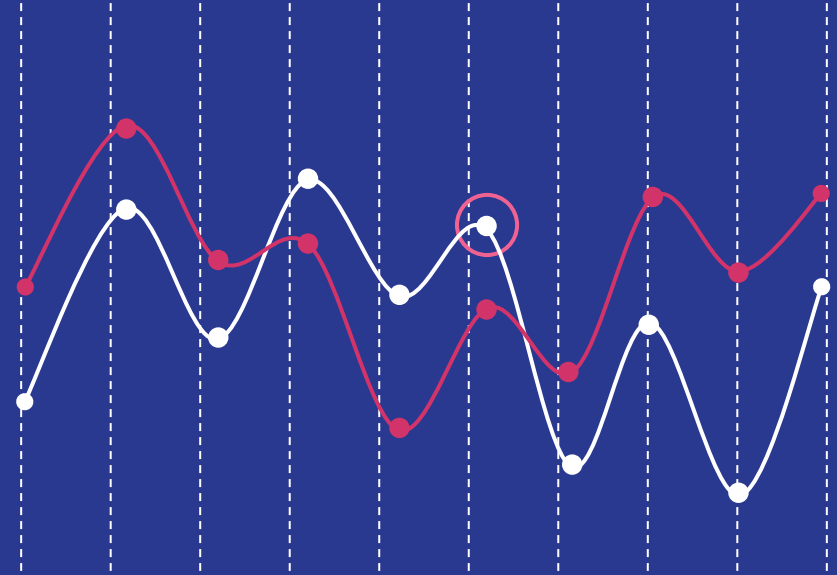
- The probability of a **type I error** is the probability of rejecting the null hypothesis when H_0 is true. Is denoted by α and is commonly referred to as the **significance level of a test**.
- The probability of a **type II error** is the probability of accepting the null hypothesis when H_1 is true. Is denoted by β and is highly affected by the sample size.
- The power of a test is defined as $1 - \beta$ or $1 - \text{probability of a type II error} = P(\text{rejecting } H_0 | H_1 \text{ true})$



Hands-on activity

R Notebook: Hands-on4_CIs in R

Lecture 6: Sample size calculation & Power analysis



Why Calculate Sample Size?

Sampling without a size calculation is a guess.

- Too small → miss a real effect (underpowered)
- Too large → waste resources, expose extra participants to risk, statistical significance not clinical
- Regulators, ethics boards, and journals expect a justification

The goal: the minimum n needed to answer your question reliably.

Inputs

Every sample size calculation requires:

Input	What it captures	You get it from...
Effect size	How big a difference matters clinically	Literature, pilot data, expert opinion
Variability (σ or p)	Spread in the outcome	Prior studies, baseline data
Significance level / Power	False positive rate / Probability to reject the null when is not true	Convention: $\alpha = 0.05$, power = 80–90%

Change inputs — the required n changes.

Effect Size: The Most Important Input

The effect size is the minimum difference worth detecting — defined clinically, not statistically.

- **Comparing two means:** $\Delta = \mu_1 - \mu_2$ (e.g., 5 mmHg difference in SBP)
- **Comparing two proportions:** $\Delta = p_1 - p_2$ (e.g., 10% difference in response rate)
- **Standardized effect size (Cohen's d):** $d = \Delta / \sigma$

Small effect size → larger n required Large effect size → smaller n required

The effect size should be the smallest difference that would change clinical practice — not the difference you expect to find.

Estimation of Mean

Estimating a population mean with desired margin of error E:

$$n = (z \cdot \sigma / E)^2$$

- $z = 1.96$ for 95% confidence
- σ = expected standard deviation
- E = acceptable margin of error

Example: $\sigma = 15 \text{ mmHg}$, $E = 3 \text{ mmHg} \rightarrow n = (1.96 \times 15 / 3)^2 = 96$

Comparing Two Means

Formula

$$n = \frac{2\sigma^2(z_{\alpha/2} + z_{\beta})^2}{\Delta^2}$$

- n = sample size **per group**
- σ = standard deviation (assumed equal in both groups)
- Δ = minimum clinically important difference ($\mu_1 - \mu_2$)
- $z_{\{\alpha/2\}} = 1.96$ for $\alpha = 0.05$ two-tailed
- $z_{\beta} = 0.84$ for 80% power; 1.28 for 90% power

Example: $\Delta = 5$ mmHg, $\sigma = 12$ mmHg, $\alpha = 0.05$, power = 80%

$n = 2 \times 144 \times (1.96 + 0.84)^2 / 25 = 91$ per group

Estimating a proportion

Estimating a proportion p with desired margin of error E :

$$n = z^2 \cdot p(1-p) / E^2$$

- Maximized when $p = 0.5$ (most conservative)
- Use known p from literature when available

Example: $p = 0.3$, $E = 0.05$, 95% CI $\rightarrow n = (1.96^2 \times 0.3 \times 0.7) / 0.05^2 = 323$

Comparing Two Proportions

$$n = \frac{(z_{\alpha/2} + z_{\beta})^2 (p_1(1 - p_1) + p_2(1 - p_2))}{\Delta^2}$$

- n = sample size **per group**
- p_1, p_2 = expected proportions in each group
- $\Delta = p_1 - p_2$

*Example: $p_1 = 0.40, p_2 = 0.25, \alpha = 0.05, \text{power} = 80\%$ * $n = (2.80)^2 \times (0.24 + 0.1875) / 0.0225 = 159$ per group

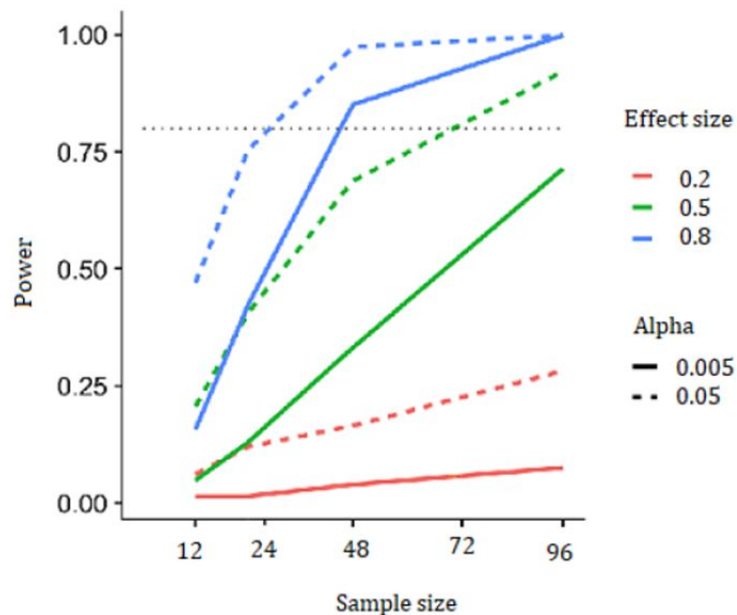
Adjustments

The formula gives the minimum n under ideal conditions. Always adjust for:

- Unequal group sizes
- Expected dropout/non-response
- Cluster design
- Multiple comparisons

Always report your assumptions. A sample size justification is only as credible as its inputs.

The Sample Size–Power Trade-off



For fixed α and effect size, power and n move together.

Going from 80% to 90% power requires ~27% more participants. Going from 80% to 95% requires ~51% more.

We will explore this trade-off interactively in the R notebook →

Web-based calculation tools

PowerAndSampleSize.com — web-based

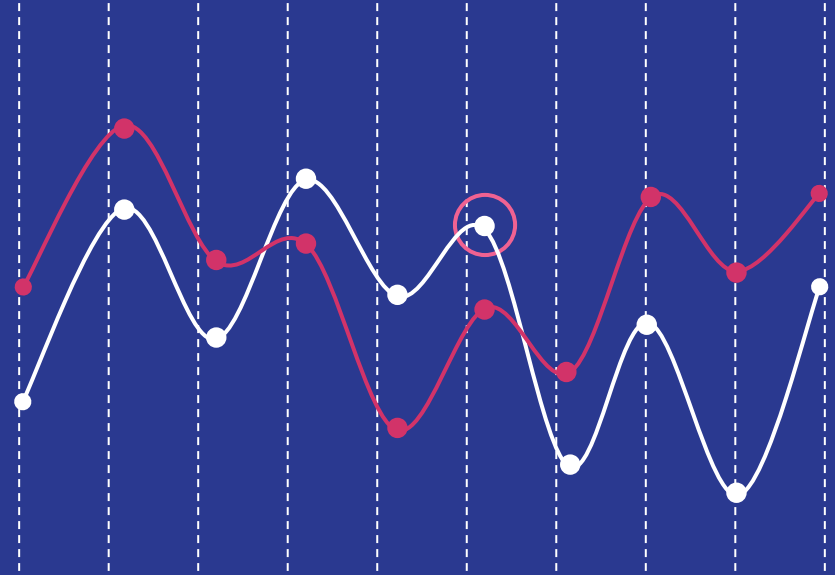
- 120+ calculators backed by peer-reviewed R packages.
- Generates shareable URLs and produces output suitable for manuscripts and grant applications.
- Integrates naturally with the `pwr` package we use in the R notebook.

Hands-on activity

R Notebook: Hands-on5_Sample size calculation in R

Lecture 7:

Parametric & Non-parametric tests

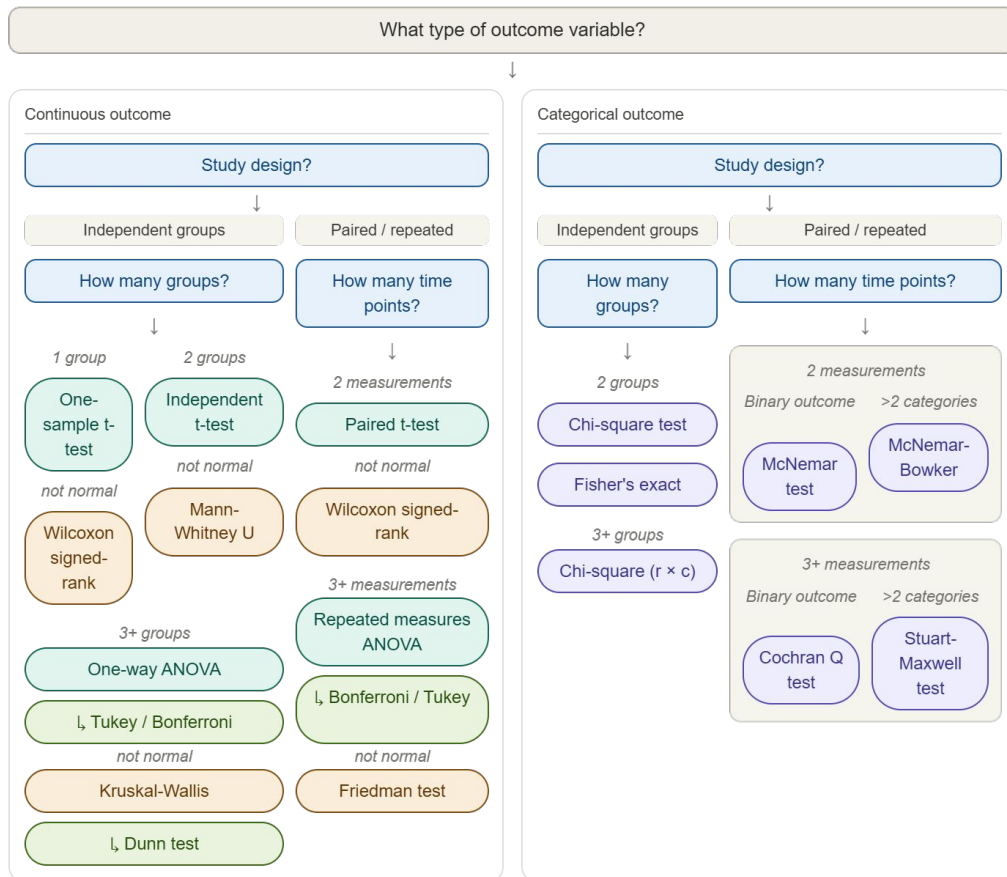


Learning objectives

- Select the appropriate test for a given research question.
- Conduct parametric and non-parametric tests in R, and extract the relevant output
- Verify the assumptions of each test using appropriate diagnostics
- Interpret p-values and confidence intervals from output

Choosing the test

- Determine the type of data
- Identify your study design



Before any parametric test — verify three assumptions:

Normality: Shapiro-Wilk, Q-Q plot · Equal variances: Levene's test · Independence: confirmed by study design

One sample t-test

1. Hypotheses (two-sided)

$$H_0: \mu = \mu_0 \text{ vs. } H_1: \mu \neq \mu_0$$

2. Test statistic

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

3. Decision

Calculate p-value (in R)

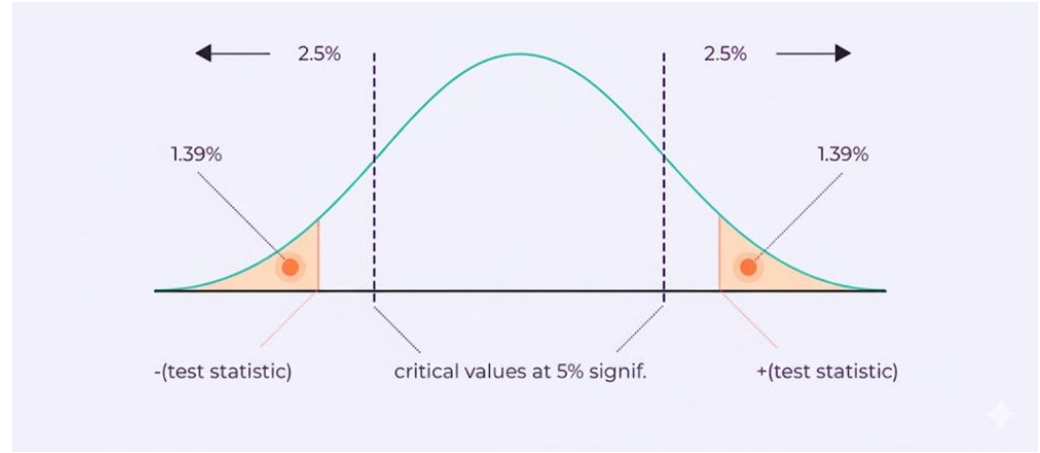
$$p = \begin{cases} 2 \times \Pr(t_{n-1} \leq t), & \text{if } t \leq 0 \\ 2 \times [1 - \Pr(t_{n-1} \leq t)], & \text{if } t > 0 \end{cases}$$

$p < 0.05$ Reject H_0 (results significant)

$P > 0.05$ Accept H_0 (results not significant)

P-value

- A p-value in statistics measures the likelihood of observing your data (or more extreme data) if the null hypothesis were true.



- A small p-value (e.g., < 0.05) suggests strong evidence against the null, implying the results aren't just random chance, while a large p-value means the data is consistent with the null hypothesis.

Paired sample t-test

1. Hypotheses

$$H_0: \Delta = 0 \text{ vs. } H_1: \Delta \neq 0$$

2. Test statistics

$$t = \frac{\bar{d}}{s_d / \sqrt{n}}$$

- Has a $t_{(n-1)}$ distribution

3. Decision

$p < 0.05$, results significant

Assumptions

1. Your dependent variable should be continuous.
2. Your independent variable should consist of two categorical, related groups.
3. The differences between pairs should be normally distributed.

Independent t-test

Assumptions

1. Your dependent variable should be measured on a continuous scale.
2. Your independent variable should consist of two categorical, independent groups (i.e., gender).
3. There should be no significant outliers.
4. Your dependent variable should be approximately normally distributed for each group of the independent variable.
5. Test the homogeneity of variances.

Independent t-test

1. Hypotheses

$$H_0: \mu_1 = \mu_2 \text{ vs. } H_1: \mu_1 \neq \mu_2$$

2. Test statistics (equal variances)

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

$$\text{where } s = \sqrt{\left[(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2 \right] / (n_1 + n_2 - 2)}$$

3 Decision

p-value < 0.05 reject null, accept H_1

- The populations means are not significantly different from each other.

Unequal variances

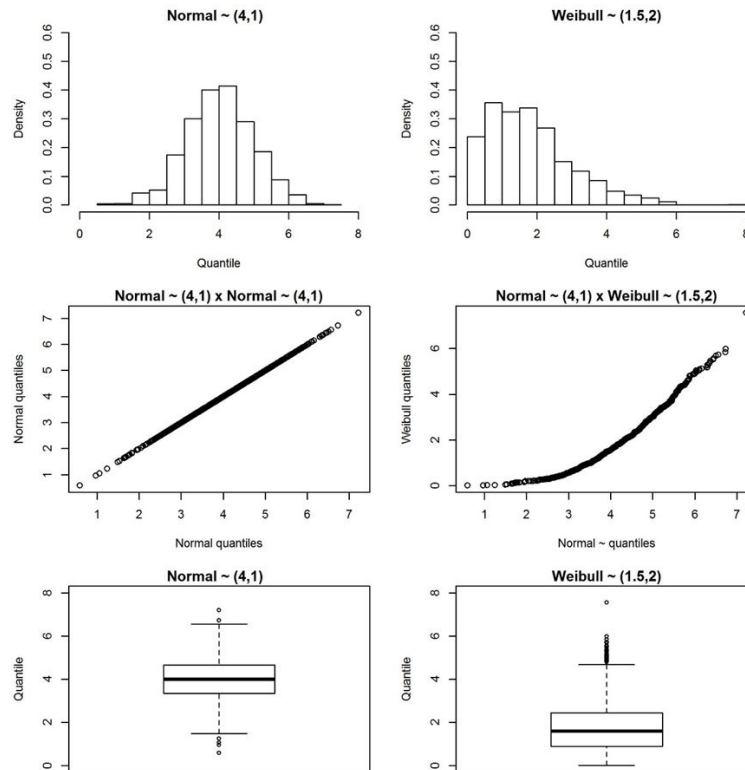
$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Compute the approximate degrees of freedom d' , where

$$d' = \frac{(s_1^2/n_1 + s_2^2/n_2)^2}{(s_1^2/n_1)^2/(n_1 - 1) + (s_2^2/n_2)^2/(n_2 - 1)}$$

Test for normality

- Graphically
 - Histogram, QQ plots, Boxplot
- Shapiro Wilk test
 - If p-value > 0.05 the data is normally distributed



Test homogeneity of variances

Levene's test

H_0 : Variances are equal

H_1 : Variances not equal

Decision

p-value > 0.05 accept H_0 – equal variances

P-value < 0.05 reject H_0 – unequal variances

One-way ANOVA

Assumptions

1. Your dependent variable should be measured on a continuous scale.
2. Your independent variable should consist of two or more categorical, independent groups.
3. There should be no significant outliers.
4. Your dependent variable should be approximately normally distributed for each group of the independent variable.
5. There is need to test homogeneity of variances.

One-way ANOVA

1. Hypotheses

$$H_0: \mu_1 = \mu_2 = \dots = \mu_a$$

H_1 : At least two means are different

2. Test statistics

$$F = s_b^2 / s_w^2$$

$\sim F_{(a-1; n-a)}$ distribution

3. Decision

$$p\text{-value} < 0.05$$

We can reject H_0 , that all the means are equal, and conclude that at least two of the means are significantly different. These results are displayed in an ANOVA table (we discuss it in R).

One-ANOVA Post-hoc analysis

If H_0 is rejected, we should perform a post-hoc analysis to see which groups are different.

We will discuss and interpret this in R

Class activity: Identify the type of test

- A hospital reports that average patient wait time in the emergency department is 45 minutes. An audit of 28 randomly selected visits finds a mean of 52 minutes (SD = 11 min). Is there evidence that actual wait times exceed the reported value? Use $\alpha = 0.05$, one-tailed.

Class activity: Identify the type of test

- A study compares haemoglobin (Hb) levels in anaemic patients randomised to oral iron supplementation vs. IV iron infusion after 12 weeks:

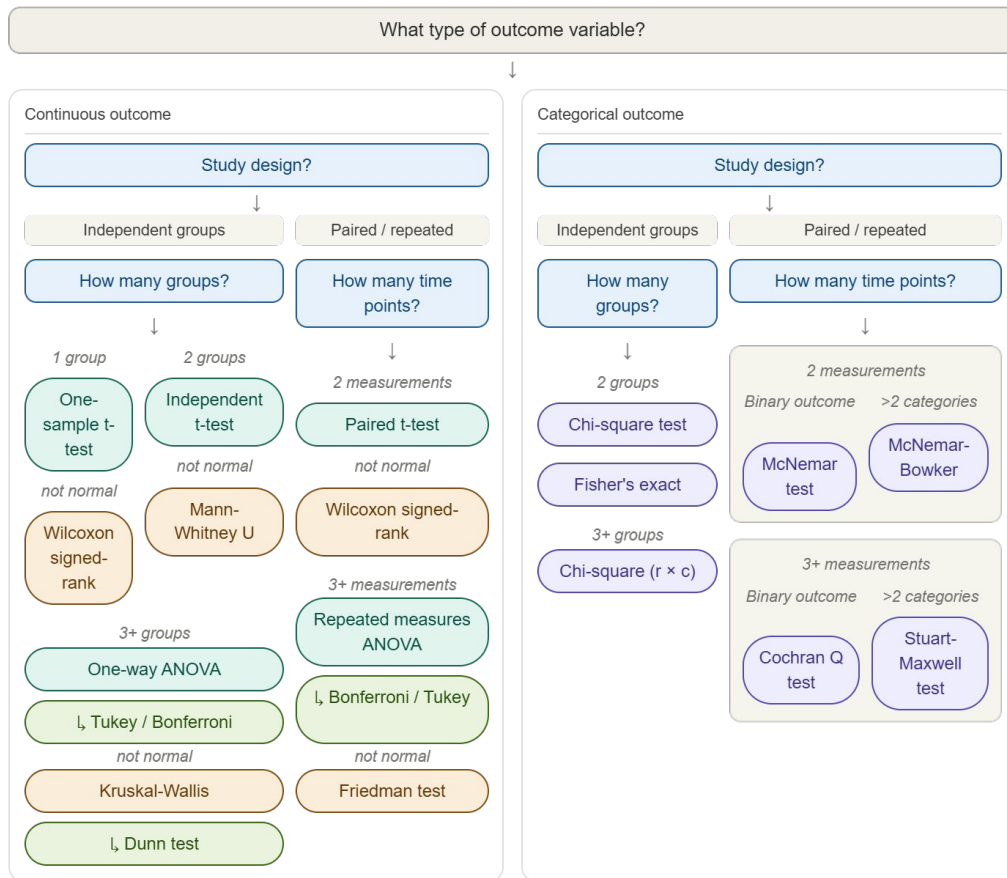
Class activity: Identify the type of test

- A clinical trial randomises 120 patients with type 2 diabetes to three treatments: metformin, lifestyle modification, or combination therapy. HbA1c (%) is measured after 6 months:

Non-parametric tests

Choosing the test

- Determine the type of data
- Identify your study design



Before any parametric test — verify three assumptions:

Normality: Shapiro-Wilk, Q-Q plot · Equal variances: Levene's test · Independence: confirmed by study design

Wilcoxon signed rank test (related samples)

H_0 : difference between the pairs follows a symmetric distribution around zero.

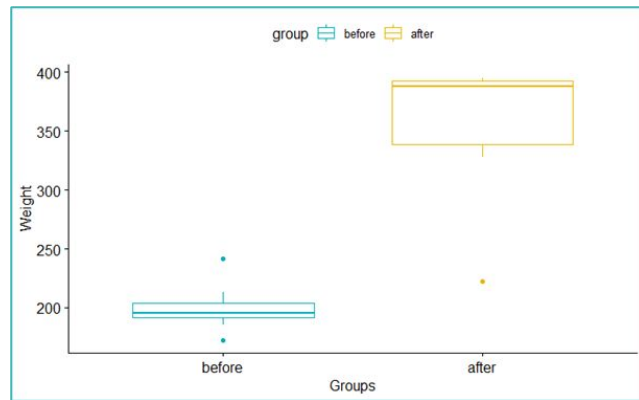
H_1 : difference between the pairs does not follow a symmetric distribution around zero.

- Test statistic

$$W = \sum_{i=1}^{N_r} [\text{sgn}(x_{2,i} - x_{1,i}) \cdot R_i]$$

- Decision

If $p < 0.05$ reject the null hypothesis (H_0)
If the difference is not symmetric around zero then there is difference between groups.



Mann Whitney U test (Independent samples)

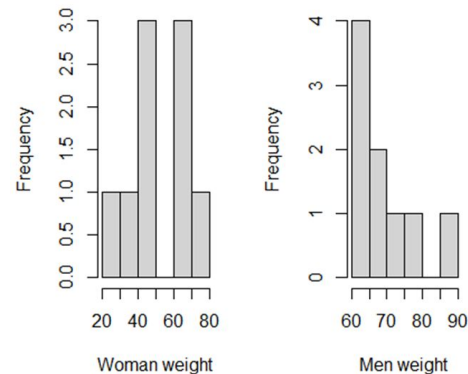
H_0 : the distributions of both populations are equal

H_1 : the distributions of populations are not equal

Test statistic

$$U = \sum_{i=1}^n \sum_{j=1}^m S(X_i, Y_j),$$

Decision based on p-value
If $p < 0.05$ reject the null hypothesis (H_0)



Kruskal Wallis (three independent groups)

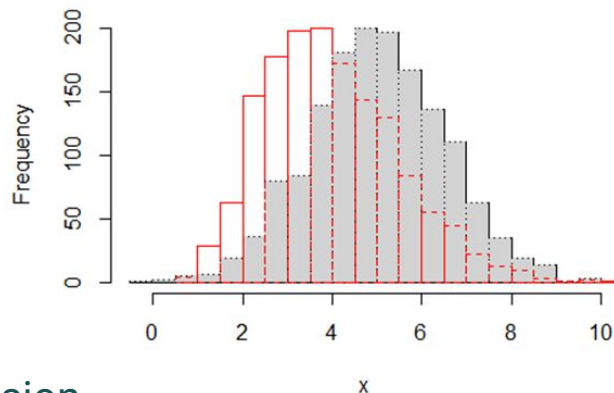
H_0 : the medians of all groups are equal,
and

H_1 : at least one population median of one
group is different from the population
median of at least one other group.

Test statistic

$$H = (N - 1) \frac{\sum_{i=1}^g n_i (\bar{r}_{i\cdot} - \bar{r})^2}{\sum_{i=1}^g \sum_{j=1}^{n_i} (r_{ij} - \bar{r})^2},$$

Under the assumption of an identically
shaped and scaled distribution for all
groups.



Decision

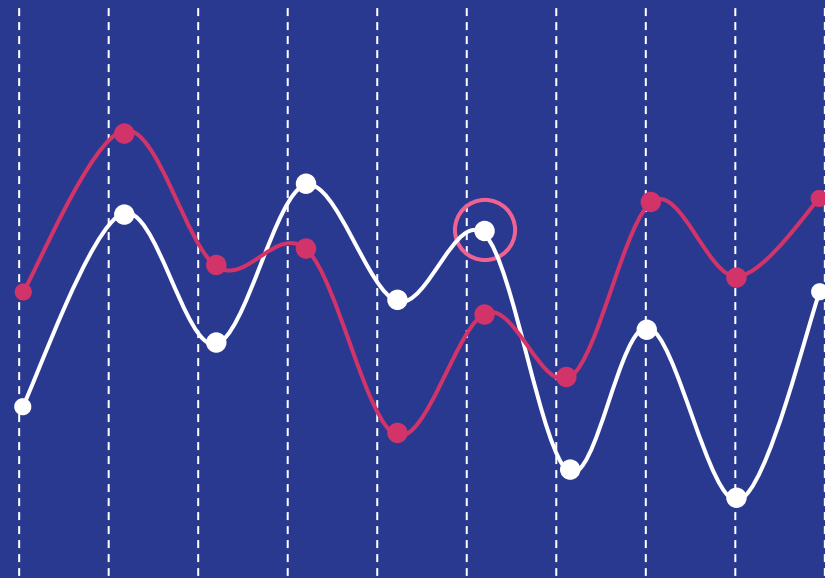
If $p < 0.05$ reject the H_0 , perform
pairwise Wilcoxon signed rank test

Hands-on activity

R Notebook:

Hands-on6_Parametric_NonparametricTests in R

Lecture 7: Linear models



Learning objectives

- One and two-way ANOVA
- Multiple linear regression
- Fitting the model
- Checking the assumptions
- Assessing model performance

The problem

We want to investigate the relationship between cholesterol levels and several predictors, including systolic blood pressure (SBP), diastolic blood pressure (DBP), age, BMI, race, and gender.

The goal is to determine how these factors influence cholesterol levels and assess the strength of their associations.

What is the purpose of fitting a model?

- To explain the relationship between the response and the predictors.
- To predict the response based on the predictors.
Often, a good model will do both.

One-way ANOVA as a linear model

It can be expressed as a **linear regression model** with categorical predictors.

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \cdots + \beta_{k-1} X_{(k-1)i} + \varepsilon_i$$

ANOVA F-test = test that all $\beta_1 = \cdots = \beta_{k-1} = 0$

Y_i :response variable

X_{gi} :dummy variables for group membership

β_0 :mean of reference group

β_g :difference between group g and reference

$$\varepsilon_i \sim N(0, \sigma^2)$$

Two-way ANOVA as a linear model

- Two-Way ANOVA evaluates effects of Factor A, Factor B, and their interaction.

Also fits naturally in a linear model framework.

$$Y_{ij} = \beta_0 + \alpha_a + \beta_b + (\alpha\beta)_{ab} + \varepsilon_{ij}$$

- Main effects: Do levels of A or B change the mean outcome?
- Interaction: Does the effect of A depend on B?
- F-tests compare each set of coefficients to 0.

Multiple Linear regression

- When there are two or more independent variables used in the regression analysis, the model is not simply linear but a multiple regression model.

The diagram shows the equation $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_p X_p + \varepsilon$ with arrows pointing to its components: Y is labeled 'Dependent Variable (Response Variable)'; β_0 is labeled 'Y intercept'; $\beta_1, \beta_2, \beta_p$ are collectively labeled 'Slope Coefficient'; X_1, X_2, X_p are collectively labeled 'Independent Variables (Predictors)'; and ε is labeled 'Error Term'.

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_p X_p + \varepsilon$$

- Y is always continuous
- The covariates X can be:
- Continuous variable (age, weight, etc.)
- Dummy variables coding a categorical covariate.

Significance and interpretation of coefficients

- The coefficients can be interpreted after testing their significance.
- If the independent variable X_i increases by one unit and all other predictors are constant, the dependent variable Y increases by β_i .

Assumptions

1. Linear relationship between the dependent and the independent variables.
2. Multicollinearity, no strong correlation between independent variables.
3. Residual values are normally distributed
4. Homoscedasticity assumes that the variance of the residual errors is similar across the values of each independent variable.

Model performance

- **Coefficient of determination R-Square**

R-squared is the proportion of the variance in the response variable that can be explained by the predictor variables.

- **Root mean squared error**

$$\text{RMSE}(\text{model, data}) = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

y_i are the actual values of the response

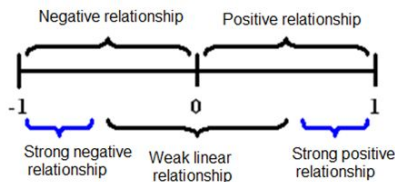
$\hat{y}_i$ (hat) are the predicted values using the fitted model

$$\left\{ \begin{array}{l} R^2 = \frac{s_{\hat{y}}^2}{s_y^2} \begin{array}{l} \text{Variance of the predicted values} \\ \text{Variance of the observed values} \end{array} \\ \\ R_{adj}^2 = 1 - (1 - R^2) \cdot \frac{n - 1}{n - p - 1} \end{array} \right.$$

Correlation

- The Pearson product-moment correlation coefficient (Pearson's correlation, for short) is a measure of the strength and direction of association that exists between two variables measured at interval scale.

$$r = \frac{1}{n-1} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{s_x} \right) \left(\frac{y_i - \bar{y}}{s_y} \right)$$



Cohen (1988):

$|r| < 0.3$ Weak

$0.3 \leq |r| < 0.5$ Medium

$|r| \geq 0.5$ Strong

Assumptions

1. Your two variables should be measured at the interval or ratio level (i.e., they are continuous).
2. There is a linear relationship between your two variables. You can check by creating a scatterplot.
3. There should be no significant outliers.
4. Your variables should be approximately normally distributed

Spearman Correlation

1. When normality assumption is violated use Spearman correlation.

$$rho = \frac{\sum (x' - m_{x'}) (y'_i - m_{y'})}{\sqrt{\sum (x' - m_{x'})^2 \sum (y' - m_{y'})^2}}$$

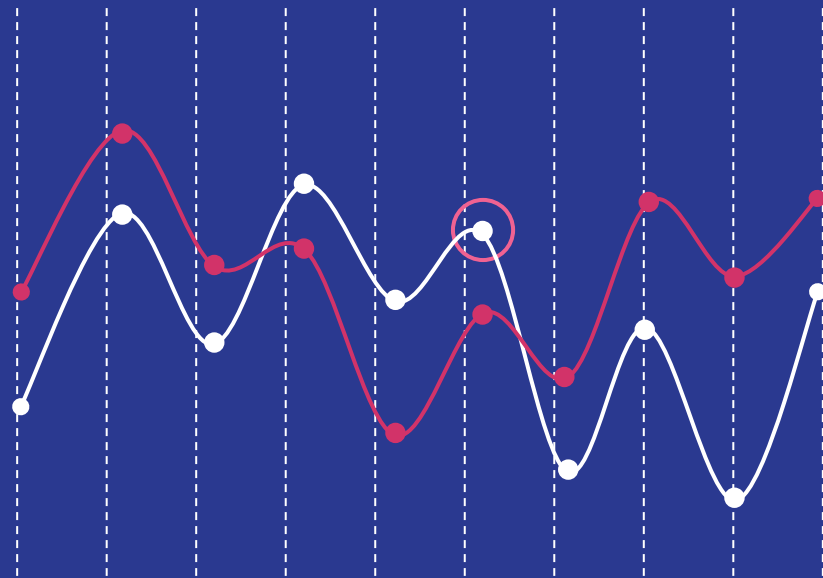
Where $x' = rank(x)$ and $y' = rank(y)$.

Hands-on activity

R Notebook: Hands-on7_Linear models in R

Lecture 8:

Categorical data analysis



Learning objectives

1. Distinguish and compute measures of association, Relative Risk and Odds Ratio.
2. Apply the Chi-Square Test for Independence and interpret results including effect size via Cramér's V .
3. Select and conduct Fisher's Exact Test, including choosing between one-sided and two-sided alternatives.
4. Evaluate goodness-of-fit for a categorical variable.

Categorical data

Data that takes a limited number of distinct values representing categories or groups — not measured on a continuous numerical scale.

- Nominal — categories with no inherent order (e.g., blood type: A, B, AB, O)
- Ordinal — categories with a natural order but unequal intervals (e.g., education level: high school < bachelor's < master's < PhD)
- Binary — special case with exactly two outcomes (yes/no, success/failure)

Key distinction: The numbers we assign to categories (e.g., 1=male, 2=female) are labels, not measurements. Arithmetic on them is meaningless.

Why Does It Require Special Methods?

Standard statistical methods assume:

- Normally distributed residuals
- Continuous, unbounded outcomes
- Equal intervals between values

Categorical outcomes violate all three. Using linear regression on a binary outcome, for example, can predict probabilities below 0 or above 1 — which is nonsensical.

Categorical data requires:

- Methods that respect probability constraints ($0 \leq p \leq 1$)
- Models appropriate for discrete, bounded outcomes

Contingency table

Cross-tabulation — summarizes the relationship between two categorical variables.

Example: Smoking status × Disease outcome

	Disease	No Disease	Total
Smoker	40	60	100
Non-smoker	10	90	100
Total	50	150	200

Key quantities from contingency tables:

- Row percentages, column percentages
- Marginal distributions
- Conditional distributions (the foundation of association analysis)

Measures of association

For 2×2 tables:

- **Relative Risk (RR):** ratio of probability of outcome in exposed group vs. unexposed group
- **Odds Ratio (OR):** ratio of the odds of the outcome in each group. Preferred when outcome is rare or study is retrospective

From the smoking example:

- $RR = (40/100) \div (10/100) = 4.0$ — smokers are 4× more likely to have the disease
- $OR = (40 \times 90) \div (60 \times 10) = 6.0$

Interpretation tip: RR is more intuitive; OR approximates RR when the outcome is rare ($< \sim 10\%$), but diverges when outcomes are common.

Chi-square test

- The chi-square test for independence, also called Pearson's chi-square test or the chi-square test of association, is used to discover if there is a relationship between two categorical variables.

Assumptions

1. Your two variables should be measured at an ordinal or nominal level (i.e., categorical data).
2. Your two variables should consist of two or more categorical, independent groups.
3. The expected counts should be larger than 5 in more than 75% of cases.

Chi-square test

Hypotheses

H_0 : Variables are independent

H_1 : Variables are related

Test statistics

$$\chi^2 = \sum [(Observed - Expected)^2 / Expected]$$

Decision

Compare χ^2 to the chi-square distribution with $(r-1)(c-1)$ degrees of freedom. Reject H_0 if $p < \alpha$.

- Variables are related.

Fisher Exact test & Goodness of fit

- **Fisher's Exact Test:** Used when sample sizes are small or expected cell counts < 5 . Computes the exact probability of observing the data (or more extreme arrangements) under H_0 .
- **Chi-Square Goodness-of-Fit Test:** Tests whether a single categorical variable follows a hypothesized distribution.

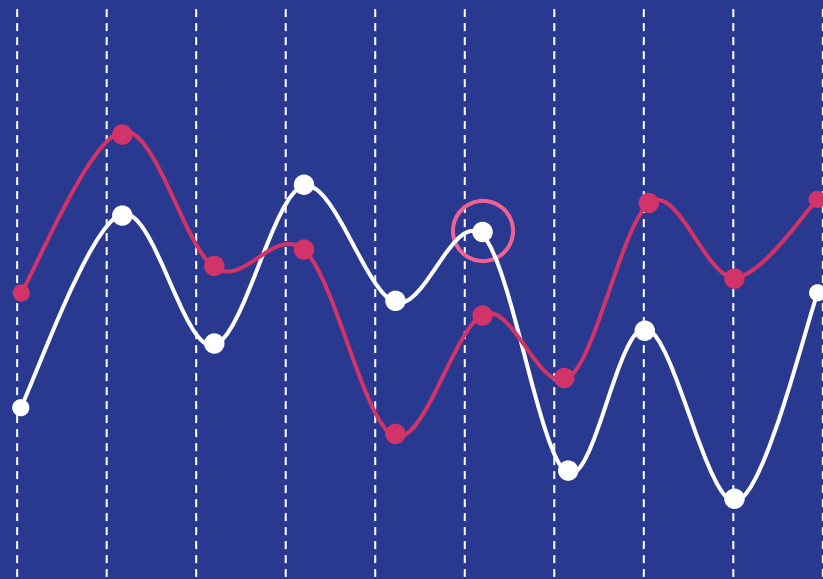
Both tests share the same logic: compare what we observe to what we'd expect under the null hypothesis.

Hands-on activity

R Notebook: Hands-on8_IntroCategoricalData_R

Lecture 9:

Generalized linear models



Why Generalized Linear Models?

Classical linear regression assumes:

- Outcome Y is continuous and unbounded
- Residuals are normally distributed with constant variance
- $E(Y) = \beta_0 + \beta_1 X_1 + \dots$ (a straight line is a sensible summary)

Many outcomes violate all three.

Why Generalized Linear Models?

Outcome type	Example	Problem with OLS
Binary	Disease yes/no	Predicts P outside $[0,1]$
Count	Hospital admissions	Predicts negative counts; variance $\neq$ mean
Proportion	% defective items	Bounded at 0 and 1
Positive continuous	Survival time	Skewed; must be > 0

The solution: Generalized Linear Models (GLMs) — a unified framework that extends linear regression to handle non-normal response distributions.

The GLM framework

1. Random component — the probability distribution of Y (from the exponential family):

- Binary outcome $\rightarrow$ Binomial
- Count outcome $\rightarrow$ Poisson
- Continuous positive $\rightarrow$ Gamma, Inverse Gaussian

2. Systematic component — the linear predictor:

$$\eta = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p$$

3. Link function $g(\cdot)$ — connects the mean $\mu = E(Y)$ to the linear predictor:

$$g(\mu) = \eta \Leftrightarrow \mu = g^{-1}(\eta)$$

Distribution	Canonical link	Name
Binomial	$\log(\mu/(1-\mu))$	Logit
Poisson	$\log(\mu)$	Log
Gaussian	μ	Identity
Gamma	$1/\mu$	Inverse

All GLMs are estimated by **Maximum Likelihood Estimation (MLE)** via iteratively reweighted least squares (IRWLS).

Logistic regression: The model

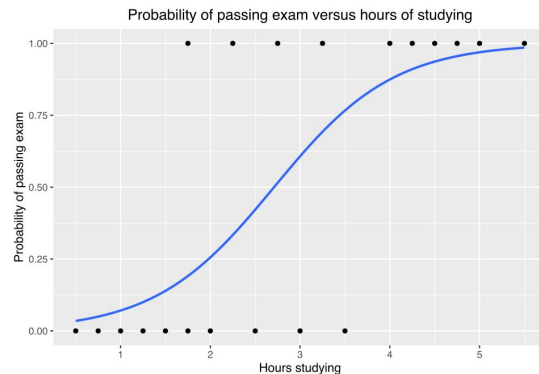
When to use: Binary outcome (0/1, yes/no, success/failure).

The model:

$$\text{logit}(p) = \ln \left(\frac{p}{1-p} \right) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_k X_k$$

The model **does not predict probability directly** but rather predicts the **log odds**, which can then be transformed back:

$$p = \frac{e^{\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_k X_k}}{1 + e^{\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_k X_k}}$$



Logistic Regression: Interpretation

The key parameter is Odds Ratio (OR)

Exponentiating a coefficient gives the **odds ratio** for a one-unit increase in that predictor, *holding all other predictors constant*:

$$OR_j = e^{\beta_j}$$

β value	OR = e^{β}	Interpretation
0	1.0	No effect on odds
0.69	2.0	Doubles the odds
-0.51	0.6	Reduces odds by 40%

Logistic Regression: Interpretation

For a continuous predictor: OR is for a 1-unit increase in X. For a categorical predictor: OR compares a category to the reference level.

Important: β coefficients are on the log-odds scale. Always exponentiate before interpreting.

Inference on coefficients:

- Wald test: $z = \hat{\beta} / SE(\hat{\beta})$ — analogous to t-test in OLS
- Likelihood Ratio Test (LRT): compares nested models via $-2 \log$ -likelihood difference; more reliable than Wald for small samples.

Logistic Regression: Interpretation

Model-level fit:

- Null deviance: fit of intercept-only model
- Residual deviance: fit of the current model
- Reduction in deviance $\approx$ explained variation (analogous to R^2)
- $AIC = -2\ell + 2p$ — for comparing non-nested models (lower = better)

Goodness-of-fit:

- Hosmer-Lemeshow test: groups observations by predicted probability deciles and tests whether observed vs. predicted counts match
- Calibration plots: predicted probability vs. observed event rate

Logistic Regression: Model Evaluation

A model can be statistically significant but practically useless. We need measures of **discrimination** and **calibration**.

Discrimination: Can the model separate cases from non-cases?

- **ROC curve:** plots Sensitivity vs. (1 – Specificity) across all classification thresholds
- **AUC (C-statistic):** probability that a randomly chosen case scores higher than a randomly chosen non-case

Classification metrics (at a chosen threshold, e.g., $p > 0.5$):

Sensitivity, Specificity, Positive Predictive Value, Negative Predictive Value, Accuracy — all derived from the confusion matrix.

AUC	Discrimination
0.50	No better than chance
0.70–0.80	Acceptable
0.80–0.90	Excellent
> 0.90	Outstanding

Poisson Regression

When to use

Outcome is a **count** — number of events occurring in a fixed time period, area, or exposure window (e.g., hospital visits per year, accidents per month, emails per day).

The Poisson distribution: $P(Y = k) = e^{(-\mu)} \mu^k / k!$ where $\mu = E(Y) = \text{Var}(Y)$

The model (log link):

$$\log(\mu) = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p$$

Equivalently:

$$\mu = e^{\beta_0 + \beta_1 X_1 + \dots + \beta_p X_p}$$

Why log link?

- Ensures predicted counts are always positive ($\mu > 0$)

Key assumption: The mean equals the variance. This is often violated in practice — leading to overdispersion.

Poisson Regression: The offset term

In many count data settings, observations differ in their **exposure** — some subjects were followed for longer, or cover a larger area.

Ignoring unequal exposure leads to **biased estimates**.

The offset: a known, fixed term added to the linear predictor with coefficient constrained to 1:

$$\log(\mu_i) = \log(t_i) + \beta_0 + \beta_1 X_1 + \dots$$

where t_i is the exposure time (or area, population size, etc.) for subject i .

- This models the **rate** (events per unit exposure) rather than the raw count:

Examples:

- Hospital length of stay →
offset = log(days admitted)
- Disease incidence →
offset = log(population at risk)

Poisson Regression: Interpretation

Interpretation: Rate Ratio (Incidence Rate Ratio)

$$IRR_j = e^{\beta_j}$$

A one-unit increase in X_j **multiplies** the expected count (or rate) by e^{β_j} holding other predictors constant.

- Same as logistic regression: MLE estimation, deviance-based fit, AIC for model selection, likelihood ratio tests for nested models, the same `glm()` interface in R.

β	$IRR = e^{\beta}$	Interpretation
0	1.0	No effect
0.41	1.5	50% more events
-0.22	0.8	20% fewer events

Poisson Regression: Overdispersion

Overdispersion: $\text{Var}(Y) > E(Y)$ — more variation than Poisson allows.

Causes: unmeasured heterogeneity, clustering, excess zeros.

Detection: Dispersion statistic $\phi = \text{Residual Deviance} / \text{df}$. If $\phi \gg 1$, overdispersion is present.

Solutions:

- Quasi-Poisson: inflates standard errors by $\sqrt{\phi}$ — same coefficients, correct inference
- Negative Binomial regression: adds an individual-level random effect; models $\text{Var}(Y) = \mu + \mu^2/\theta$
- Zero-inflated models: when zeros are more frequent than Poisson predicts

GLMs: Model selection

Model building strategies:

- Likelihood Ratio Test (LRT): test whether adding a predictor significantly improves fit: $-2(\ell_{\text{reduced}} - \ell_{\text{full}}) \sim \chi^2(\text{df})$
- AIC / BIC: lower is better; BIC penalizes complexity more heavily — useful when parsimony matters
- Stepwise selection: automated but use with caution — can overfit and doesn't account for multiple testing

GLMs: Diagnostics

Issue	How to detect	Fix
Influential observations	Cook's distance, leverage	Investigate outliers
Complete separation (logistic)	Coefficients blow up to $\pm\infty$	Penalized regression (Firth)
Overdispersion (Poisson)	Deviance/df $\gg 1$	Quasi-Poisson or Negative Binomial
Multicollinearity	VIF > 10	Drop/combine predictors
Non-linear effects	Residual plots, splines	Add polynomial or spline terms

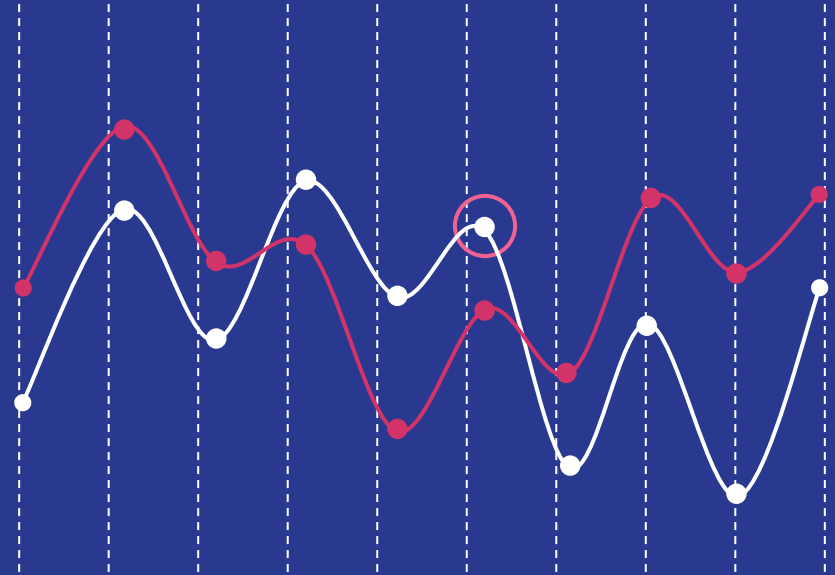
Residuals in GLMs

- **Pearson residuals:** $(\text{Observed} - \text{Fitted}) / \sqrt{\text{Var}(\text{Fitted})}$
- **Deviance residuals:** contribution of each observation to the total deviance
- Plot both against fitted values and predictors to check for patterns.

Hands-on activity

R Notebook: Hands-on9_glmLogisticPoisson_R

Lecture 10: Intro to survival analysis



What Is Survival Analysis?

Survival analysis is a branch of statistics concerned with **time until an event occurs** — not just whether the event happens, but *when*.

The event (also called the "failure") can be:

- Death (the origin of the name)
- Disease onset or diagnosis
- Hospital readmission
- Recovery or remission
- Relapse after treatment
- Device failure or drug discontinuation

What Is Survival Analysis?

The time variable is the duration from a defined starting point (study entry, diagnosis, surgery, randomization) to the event or to the end of observation.

Why not just use means or proportions?

Because medical studies almost always end before every participant has experienced the event. We need methods that extract information from both complete and incomplete observations.

The Problem of Censoring

Censoring is the defining challenge of survival data. It occurs when we know a subject survived beyond a certain time, but do not know their exact event time.

Right censoring (most common):

- The study ends before the event occurs
- The patient is lost to follow-up
- The patient withdraws or dies of an unrelated cause

Left censoring: The event occurred before observation began (e.g., the patient was already infected at enrollment).

Interval censoring: We know the event occurred within a time window but not exactly when (e.g., detected at a scheduled clinic visit).

The Problem of Censoring

Key assumption — non-informative censoring: Censored subjects must have the same underlying survival prospects as those who remain under observation. If sicker patients are more likely to drop out, this assumption is violated and estimates will be biased.

Ignoring censored subjects (complete-case analysis) is almost always **wrong** — it discards valid information and systematically overestimates survival.

Key Concepts and Functions

1. Survival function $S(t)$

$$S(t) = P(T > t)$$

The probability of surviving (not experiencing the event) beyond time t . $S(0) = 1$; $S(\infty) = 0$.

2. Hazard function $h(t)$

$$h(t) = \lim_{\Delta t \rightarrow 0} \frac{P(t \leq T < t + \Delta t \mid T \geq t)}{\Delta t}$$

The instantaneous rate of the event at time t , given survival up to t . Not a probability — can exceed 1.

3. Cumulative hazard $H(t)$

$$H(t) = \int_0^t h(u) du$$

4. Relationship between $S(t)$ and $h(t)$

$$S(t) = \exp\{-H(t)\}$$

- These four functions contain the same information — knowing any one determines the others. The choice of which to model is one of the key decisions in survival analysis.

Visualizing Survival: the Kaplan-Meier Estimator

The **Kaplan-Meier (KM) estimator** is the standard non-parametric method for estimating the survival function directly from data, without assuming any underlying distribution.

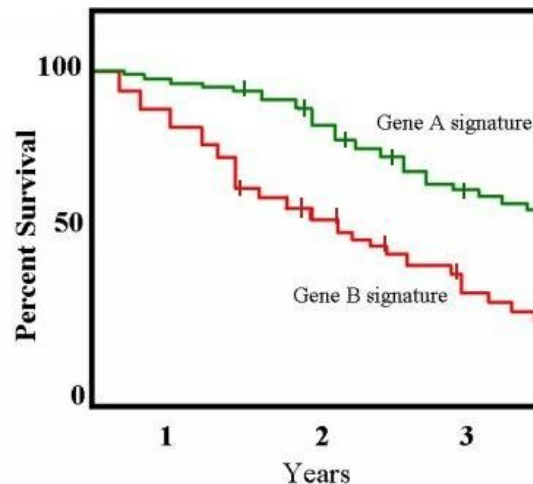
The formula:

$$\hat{S}(t) = \prod_{t_i \leq t} \left(1 - \frac{d_i}{n_i} \right)$$

where at each event time t_i : d_i = number of events, n_i = number at risk just before t_i .

How it works:

- The curve starts at $S(0) = 1$
- It drops only at observed event times (not at censoring times)
- Censored observations are shown as tick marks on the curve
- Between event times, the curve is flat (a step function)



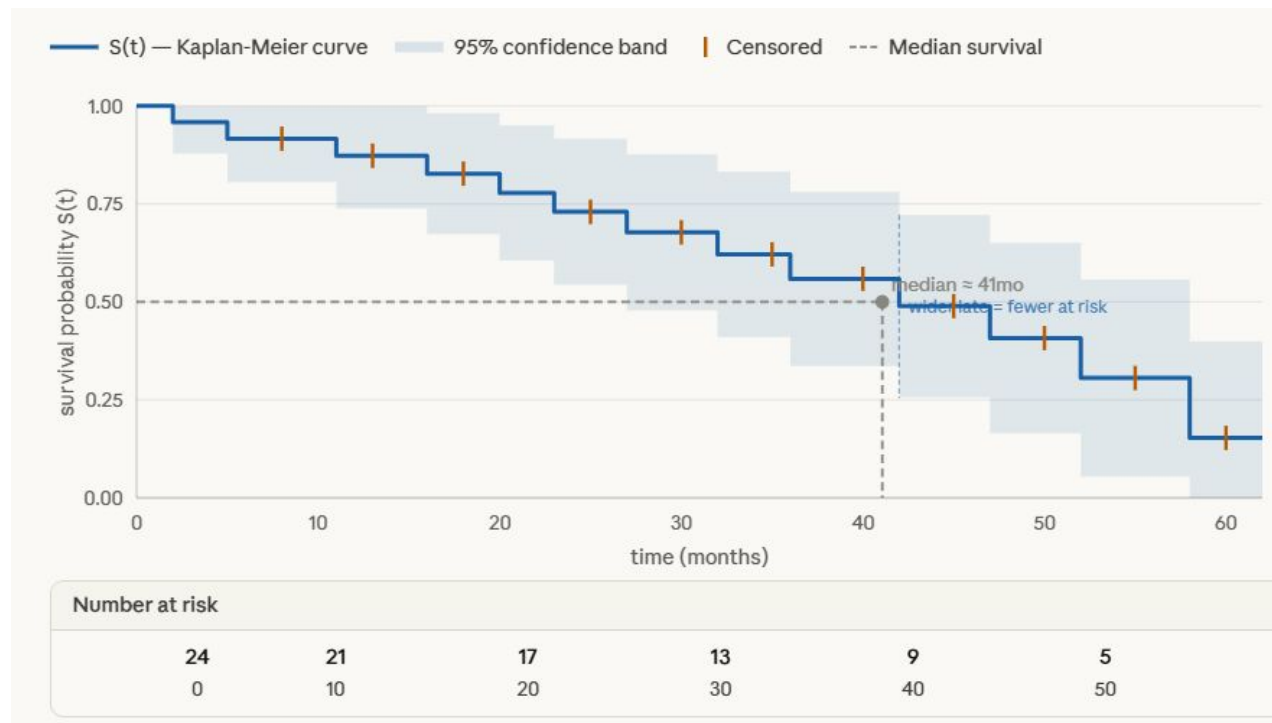
Visualizing Survival: the Kaplan-Meier Estimator

The KM curve is:

- The first thing to plot in any survival analysis
- The reference against which parametric models are judged
- Essential for communicating results to clinicians

Confidence intervals are typically computed on the log or log-log transformed scale for better coverage near 0 and 1.

Reading a KM Plot



Comparing Groups: The Log-Rank Test

The **log-rank test** is the standard method for testing whether two or more survival curves are statistically different.

Hypotheses:

$H_0: S_1(t) = S_2(t)$ for all t

H_1 : The survival functions differ at some time t

Mechanics: At each event time, the test compares the *observed* number of events in each group to the *expected* number if the null were true. These discrepancies are summed across all event times into a chi-square test statistic.

Comparing Groups: The Log-Rank Test

Properties

- Non-parametric — makes no distributional assumptions
- Most powerful when the hazard ratio between groups is constant over time (proportional hazards)
- Loses power when hazard functions cross (e.g., one treatment helps early, the other helps late)

Weighted log-rank tests

- **Wilcoxon (Breslow) test:** Weights by risk set size — gives more weight to early differences
- **Fleming-Harrington:** Family of weights for emphasizing early or late differences

Limitation: The log-rank test answers "is there a difference?" not "how large is the difference?" — for the latter, we need the hazard ratio from a Cox model.

The Hazard Ratio

The **hazard ratio (HR)** is the primary effect measure in survival analysis — analogous to the odds ratio in logistic regression.

$$HR = h_1(t)/h_2(t)$$

Under the **proportional hazards assumption**, this ratio is constant over time — so we estimate a single number summarizing the relative risk at any point.

Interpretation:

- $HR = 1.0 \rightarrow$ no difference in hazard between groups
- $HR = 0.5 \rightarrow$ treated group has half the instantaneous event rate of control at any time (protective)
- $HR = 2.0 \rightarrow$ treated group has twice the event rate (harmful)

The Hazard Ratio

Example

If the HR for a chemotherapy drug vs. placebo is 0.65:

"Patients receiving the drug had a 35% lower rate of death at any given time compared to placebo, after adjusting for covariates."

Important distinction from relative risk

- Relative risk compares cumulative probabilities: $P(\text{event by time } t)$ in group 1 vs. 2
- Hazard ratio compares instantaneous rates conditional on survival to time t
- They are numerically different and answer different questions

The HR is the output of the Cox proportional hazards model — the dominant regression framework in survival analysis.

The Cox Proportional Hazards Model

The Cox model (Cox, 1972) is the most widely used regression model in medical survival analysis. It relates covariates to the hazard function while making minimal distributional assumptions.

The model:

$$h(t | X) = h_0(t) \cdot \exp(\beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p)$$

Key components

- $h_0(t)$: the baseline hazard — the hazard when all covariates equal zero. Left completely unspecified (non-parametric component).
- $\exp(\beta_j X_j)$: the parametric component — multiplicative effect of each covariate on the hazard.

The Cox Proportional Hazards Model

- **Why it's called "semi-parametric"**

The parametric part (the β coefficients $\rightarrow$ hazard ratios) is estimated without specifying the shape of $h_0(t)$.

- **Estimation**

Partial likelihood (Cox, 1975) — a clever trick that eliminates $h_0(t)$ from the likelihood, allowing β estimation using only the ordering of event times, not their actual values.

- **The proportional hazards assumption**

All covariates shift the baseline hazard by a constant multiplicative factor.
This must be tested — it is not guaranteed to hold.

How to check assumption?

Method	What to look for
Log-log plot	Plot $\ln(-\ln(S(t)))$ vs. $\ln(t)$ for each group; parallel lines = assumption holds
Schoenfeld residuals	Plot against time; a horizontal band (no trend) = assumption holds
Time-varying coefficients	Add an interaction $\beta(t) \times \text{time}$; significant coefficient = violation
Graphical KM overlay	Plot observed KM vs. predicted survival; close agreement = good fit

What to do when the assumption is violated?

- **Stratification:** Allow different baseline hazards for each group; sacrifice estimation of the stratifying variable's effect
- **Time-varying coefficients:** Let β change over time (e.g., β_1 for $t < 1$ year, β_2 for $t \geq 1$ year)
- **Parametric models:** Specify a flexible hazard shape (e.g., Weibull, piecewise exponential)
- **Restricted mean survival time (RMST):** Model-free summary that avoids the assumption entirely

Parametric Survival Models

- When we are willing to specify the shape of the baseline hazard, **parametric models** offer greater efficiency and allow unconditional survival time predictions.

Distribution	Hazard shape	Typical use
Exponential	Constant (flat)	Simple baseline; memoryless property
Weibull	Monotone increasing or decreasing	Most common parametric choice; generalizes exponential
Log-normal	Increases then decreases (humped)	Cancer recurrence, infection incubation
Log-logistic	Increases then decreases	Similar to log-normal; closed-form $S(t)$
Gompertz	Exponentially increasing	Mortality in older populations

Practical Issues: Study design

- **Entry time:** Use a consistent, clinically meaningful starting point (date of diagnosis, randomization, surgery). Mixing entry times introduces bias.
- **Administrative censoring:** Subjects still event-free at study close are censored at the closing date — this is informative only if closing date is related to prognosis (usually it isn't).
- **Competing risks:** A patient who dies of cardiovascular disease cannot later develop the cancer endpoint of interest. Standard methods treat competing events as censoring — this overestimates the cumulative incidence.
Cause-specific hazards or the **Fine-Gray subdistribution model** are the correct approaches.

Practical Issues: Sample size

- Rule of thumb: **10–15 events per covariate** in the Cox model (EPE — events per parameter)
- Below this threshold, coefficient estimates are unstable and confidence intervals are unreliable
- Power depends on the number of *events*, not the total sample size

Reporting standards

- Always report KM curves with risk tables
- Report HRs with 95% CIs and p-values
- State explicitly whether and how the PH assumption was tested
- Provide median survival with 95% CI for each group

Summary & Key Takeaways

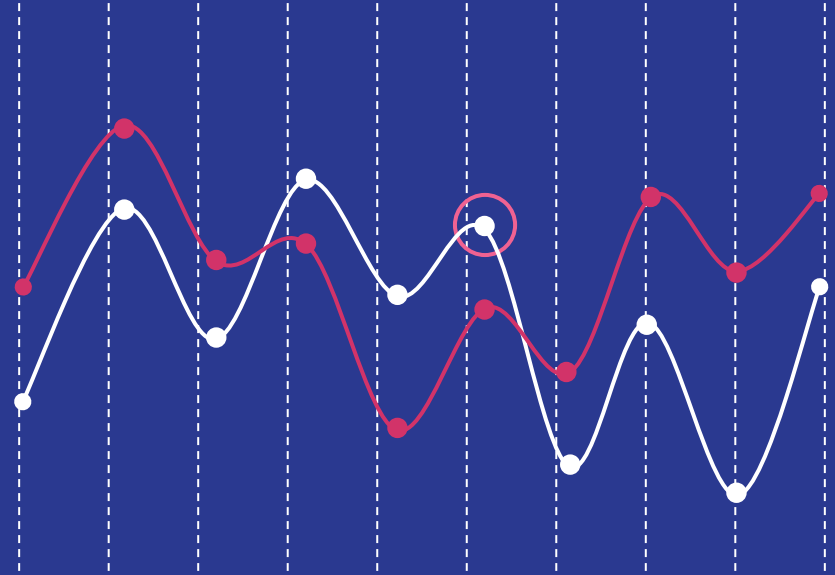
Three things never to forget

- Characterize your censoring mechanism — is it plausibly non-informative?
- Check the proportional hazards assumption — it is an assumption, not a fact
- Report risk tables — reliability of the KM curve deteriorates as the risk set shrinks

Essential tools

- Kaplan-Meier curve — describe and visualize survival; always the starting point
- Log-rank test — test whether two or more survival curves differ
- Hazard ratio — quantify the magnitude and direction of a difference
- Cox proportional hazards model — adjust for confounders; the workhorse of survival regression

Lecture 11: Documentation & Reproducibility



Why This Lecture

Statistical analysis in medical research is only as valuable as its communication and reproducibility.

— Poor reporting

- Methods sections that omit which software, which version, or which test was used
- Results sections that report p-values without effect sizes or confidence intervals
- Tables with unlabelled columns, missing units, or undefined abbreviations
- Conclusions that go beyond what the analysis actually shows

Why This Lecture

— Irreproducibility

- A 2016 survey (Baker, *Nature*) found that more than 70% of researchers had tried and failed to reproduce another scientist's results
- A 2015 Reproducibility Project (*Science*) successfully replicated only 36 of 100 published psychology studies
- Medical research faces the same crisis — with higher stakes

The Statistical Methods Section: What Must Be There

The methods section must allow a statistically competent reader to reproduce your analysis and a clinician reader to evaluate its appropriateness.

The non-negotiables

1. Study design and unit of analysis — what kind of study is this, and what does each row of data represent?
1. Outcome variables — primary and secondary outcomes, how they were defined and measured
2. Explanatory variables — which covariates were included and why (a priori specification vs. data-driven selection must be distinguished)
3. Statistical tests and models — name every method used; don't write "appropriate statistical tests were used"

The Statistical Methods Section: What Must Be There

The non-negotiables

5. Assumptions and how they were checked — normality, proportional hazards, overdispersion, etc.
6. Missing data strategy — complete-case, multiple imputation, or other; not mentioning this is a red flag
7. Significance threshold and multiple testing — pre-specified α , Bonferroni correction, FDR, or none
8. Software and version — "R version 4.3.1 (2023-06-16), using the survival package version 3.5-7"

The Methods Section: What to Avoid

 Wrong	 Right
"Data were analysed using R"	"Analyses were conducted in R v4.3.1; Cox models were fitted using the survival package v3.5-7"
"Appropriate statistical tests were applied"	"Group differences were tested using the Wilcoxon rank-sum test (continuous) and Fisher's exact test (categorical)"
"Normality was assumed"	"Normality of residuals was assessed using the Shapiro-Wilk test and Q-Q plots; log-transformation was applied to bilirubin due to right skew"
"Variables were selected based on significance"	"Covariates were pre-specified based on clinical knowledge; age, sex, and baseline severity were included regardless of p-value"
"Missing data were excluded"	"Missing data occurred in 8.3% of observations, primarily in weight loss (14%). Multiple imputation by chained equations (MICE, m = 20) was used under a MAR assumption"
"P < 0.05 was considered significant" alone	Add: "No adjustment for multiple comparisons was made; all secondary analyses are exploratory"

The Results Section

A p-value alone tells you almost nothing clinically useful. Always pair statistical significance with the size and precision of the effect.

Analysis	Effect measure	Report as
Two-group comparison	Mean difference	MD = 2.3 (95% CI: 0.8–3.9)
Binary outcome	Odds ratio / Risk ratio	OR = 1.74 (95% CI: 1.21–2.50)
Survival analysis	Hazard ratio	HR = 0.65 (95% CI: 0.48–0.88)
Correlation	Pearson's r or Spearman's ρ	$r = 0.42$ (95% CI: 0.28–0.54)
Categorical association	Cramér's V	$V = 0.31$
Regression	Standardised β or partial R^2	$\beta = 0.31$ per SD increase

The Results Section

The language of uncertainty

- "The hazard ratio was 0.65 (95% CI: 0.48–0.88), indicating a statistically significant 35% reduction in the hazard of death."
- Never: "The treatment was highly significant ($p = 0.003$)."
- Always ask: *Would this effect size matter to a clinician or patient if it were real?*

Statistical significance $\neq$ clinical significance. With large samples, trivially small effects reach $p < 0.05$. Report both.

The results section

Table — Baseline characteristics

- Rows = variables, Columns = groups (or All participants)
- Report mean (SD) for symmetric continuous variables; median (IQR) for skewed
- Report n (%) for categorical variables
- Include total N in each column header
- Do not include p-values in RCT Table 1 (they are meaningless — imbalance is by chance)

Results tables

- Every table needs a title that stands alone without reading the text
- Define all abbreviations in a footnote
- Use consistent decimal places within a column
- Report 95% CIs in every regression table — not just p-values

The results section

Figures

- Kaplan-Meier plots must include risk tables
- Box plots should show the actual data (jittered points) for $n < 50$
- Axis labels must include units
- Color should not be the only way to distinguish groups (for colour-blind accessibility)
- Every figure needs a caption that is self-explanatory

General rule: A reader should be able to understand each table or figure without reading the surrounding text.

Reproducibility

Reproducibility (computational): Given the same data and the same code, a different analyst obtains the same numerical results.

Replicability (scientific): A new study using new data reaches the same scientific conclusions.

This lecture focuses on **computational reproducibility** — the minimum standard for any quantitative research.

Reproducibility

The goal for this course: Reach at least *Reproducible* for every piece of analysis you submit. Aim for *Robustly reproducible*.

The reproducibility spectrum

Level	Description
Not reproducible	Analysis done interactively; no code saved; data modified in Excel
Weakly reproducible	Code exists but requires manual steps; paths are hardcoded; order matters
Reproducible	Code runs top-to-bottom on a fresh session and produces all reported outputs
Robustly reproducible	Code + environment + data are version-controlled and archived
Openly reproducible	Code + data deposited in a public repository with a persistent identifier (DOI)

The Reproducible Workflow: Core Principles

Principle 1 — Separate raw data from everything else Raw data is read-only. Never overwrite it, never edit it manually. All transformations happen in code.

Principle 2 — One project, one directory All files belonging to an analysis live under one root folder. Use relative paths, never absolute paths (`./data/` not `/Users/yourname/Desktop/old stuff/analysis3/data/`).

Principle 3 — Scripts flow in one direction `01_import.R` → `02_clean.R` → `03_analyse.R` → `04_report.Rmd`. Each script reads from a saved intermediate file; no script depends on objects that exist only in memory from a previous session.

Principle 4 — Set a random seed Any analysis involving simulation, resampling, or multiple imputation must set `set.seed()` before the relevant step. Document the seed in both the code and the methods section.

Principle 5 — Never do data manipulation in the reporting file The `.Rmd` or `.qmd` file should call pre-processed data and run analyses — not reshape, recode, or filter interactively.

Principle 6 — Version control everything Git tracks every change to every file. You never lose work, and you can always return to the state of the code at the time of submission.

Project Structure and File Organisation

The RStudio Project file (`.Rproj`) sets the working directory automatically for every collaborator — no more `setwd()`.

The `renv` package records the exact version of every R package used. Any collaborator (or future-you) runs `renv::restore()` and gets the identical package environment.

Naming conventions:

- Use `snake_case` for file and object names
- Prefix scripts with numbers for ordering: `01_`, `02_`, ...
- Date-stamp outputs if running multiple versions: `table1_2024-11-15.csv`
- Never: `final.R`, `final_v2.R`, `final_ACTUAL_final.R`

R Markdown and Quarto: Literate Programming

Literate programming weaves code, output, and narrative into a single document. The reported numbers are generated by the code — they cannot be copied incorrectly.

manuscript.Rmd

- YAML header → output format, title, author, date
- Text chunks → narrative written in Markdown
- Code chunks → R code that runs and produces output inline
 - {r setup} → global options, library() calls
 - {r table1} → generates Table 1 directly in the document
 - {r km-plot} → generates the KM figure
 - {r cox-model} → runs Cox model, extracts HR, CI, p
- Inline code → ``r round(hr, 2)`` embeds the value in text

R Markdown and Quarto: Literate Programming

The critical advantage of inline code:

```
"The hazard ratio was `r round(exp(coef(cox1)["sexFemale"]), 2)`  
(95% CI: `r round(ci[1], 2)`–`r round(ci[2], 2)`), p = `r round(pval, 3)`."
```

When the model changes, every number in the text updates automatically. There is no copy-paste step where a transcription error can occur.

Quarto is the next-generation successor to R Markdown — supports R, Python, Julia, and Observable JS in one document.

Data Integrity: Protection from the Source

Data integrity means that the values in your analysis dataset accurately reflect what was measured or recorded — and that you can demonstrate this.

At the point of data collection:

- Use validated electronic data capture (REDCap, OpenClinica) with built-in range checks
- Define allowable ranges for every variable before data collection begins
- Record date and time of entry; flag amendments with reason codes

At the point of analysis:

- Document every data cleaning decision in code (not in Excel, not in memory)
- Write an **audit trail script**: starting from the raw file, reproduce every derived variable and exclusion
- Cross-check key variables against source documents on a random sample
- Version-control the cleaned dataset with a hash (MD5/SHA) so you can prove it has not been modified

The golden rule: If you cannot reconstruct your analysis dataset from the raw data using only your scripts, your data integrity is not demonstrable.

Summary

Before submitting any statistical analysis:

Methods section ✓

- Named every statistical test and model used
- Specified software, version, and key packages
- Described how assumptions were checked
- Stated the missing data strategy
- Pre-specified primary vs. exploratory analyses
- Declared significance threshold and multiple-testing policy

Reproducibility ✓

- All analysis in scripts, not interactive sessions
- Raw data is untouched; all cleaning is in code
- `set.seed()` before any stochastic step
- Used RStudio Projects and relative paths
- Package versions recorded (`renv` or `sessionInfo()`)
- Report generated from R Markdown / Quarto

Results section ✓

- Reported sample size with exclusions explained
- Presented Table 1 with appropriate summary statistics
- Reported effect sizes and confidence intervals (not p-values alone)
- Labelled all tables and figures with self-contained captions
- Distinguished primary from secondary results

Resources

- EQUATOR Network (equator-network.org) — reporting guidelines (CONSORT, STROBE, TRIPOD)
- Stodden et al. (2016) — *Science* — "Enhancing reproducibility for computational methods"
- Bryan, J. — "Excuse me, do you have a moment to talk about version control?" — *The American Statistician*
- Hadley Wickham — *R for Data Science* (free at r4ds.had.co.nz)